Frustrated connections on a substitution tiling: a preregistered unification programme and its scorecard

Status report — not a theory of everything

[TODO: author list and affiliations (placeholder)]

Draft v0.6, 7 November 2026 (internal; not submitted)

Abstract

We study one mathematical object: a group-valued connection on the panel complex of the three-dimensional chair substitution tiling, together with the bottom ("floor") of its covariant Laplacian $H_U = 24 - A_U$. We collect what is proved, certified, simulated, conjectured, analogised and falsified about it in a labelled ledger, and record preregistered attempts to move it toward physics, failures included. The strongest results are mathematical and lattice-level. For any finite-group Wilson ensemble on the hull the infinite-volume floor is 0 almost surely at every coupling, so the floor is not an order parameter [P, hand proof, unverified]. The Wilson theory on the chair complex $\times\mathbb{Z}$ has a positive transfer matrix, by the standard construction [P, hand proof, unverified]. For the θ-deformation of the frame connection, the second-order floor coefficient c_L converges, for every primitive frame rule, to a corrector formula; this is an application of known ergodic-homogenisation results to the framed-chair hull [P, hand proof, unverified]. An exact enumeration over all 6^8 rules gives $\liminf c_L \geq 3527/1012$ [C], conditional on two of the hand-proved theorems. Numerically, along $\theta = \kappa a$ and for supertile levels $L \leq 6$, the second-order coefficient behaves like a mass term [E]; the continuum limit itself is [X]. Six of the seven physics requirements of our scorecard remain open: gauge group, chirality, generations, Lorentzian 3+1 spacetime with a continuum limit, gravity, and a derived physical constant. Chirality, generations and gravity carry recorded obstructions. Five Phase 3 preregistered predictions (G1 H1, H2, H3b; G2 W3c, B4) were falsified; these and earlier [F] items are reported. Nothing here is a theory of everything, and nothing here is observational.

Labels. Every claim carries exactly one of six fixed labels. The label printed is taken from the source and may be weaker than it, never stronger: [P] proved, [C] computed/certified, [E] numerical/simulated evidence, [X] conjecture, [A] analogy, [F] falsified. Where the source uses a compound label (a [P] reduction plus a [C] computation) the paper prints the weaker component, [C], and keeps the source label in the evidence line. Two claims carry non-standard labels in the source; they are printed no stronger than [E] (or [X] when there is no numerical evidence) with the marker [label pending author ruling]. Every claim whose printed label differs from its source label is listed with the reason in Appendix C. A claim that depends on a hand proof inherits that dependence, which is stated in the claim. Hand proofs that are neither machine-checked nor externally reviewed carry unverified. Everything labelled [E] is simulated (Monte Carlo or floating-point spectra on finite patches); none of it is observational data. Appendix A lists every labelled claim.

1 Introduction

This note reports a programme built around a single mathematical object: a group-valued connection on the panel complex of the three-dimensional chair substitution tiling, and the bottom of its covariant

Laplacian. Several bodies of the author's earlier work (Chair44, a long series of computational phases on the chair tiling; the QIS and p_vs_np_repo repositories; and a set of geometric-unity notes; Section 2) turn out to share this object, or to fail to share it in ways that can be stated precisely. The word "unification" is meant in this narrow sense: one object, one ledger of every claim made about it, each with one of six labels, and a sequence of preregistered attempts to move the object toward physics, reported whatever their outcome.

The title's disclaimer is a rule, not modesty. We would call the programme a theory of everything only if every row of the physics scorecard of Section 5 were [P] or [C]; six of the seven rows are open (TOE_ATTEMPT.md l.3).

Section 2 says what was read and what was excluded. Section 3 defines the object and lists what is proved or certified about it. Section 4 reports the six Phase 3 gaps, each with its frozen predictions and their scores, and the later proof of the $\theta \to 0$ limit. Section 5 gives the scorecard, Section 6 the refutation routes, Section 7 the relation to prior work (including the homogenisation literature behind Theorem F), and Section 8 a summary. Appendices list every labelled claim, the flags against the ledger, the label changes made relative to the sources, and the reproducibility record.

2 Sources and what was excluded

The work underlying this note was read as data from the author's repositories: a ledger of 1243 claims from Phase 0, 37 entries from Phase 3 and 8 from PROOF.md (45 ids `TOE-...`), 1288 in total (TOE_ATTEMPT.md §1 l.15).[1] Breakdown: Chair44 570, QIS 423, p_vs_np_repo 243, geometric-unity notes 5, external 2 (TOE_ATTEMPT.md l.17–21). Of these, the two "external" entries are third-party claims recorded and not checked (`I-UG-EXT` below); the other repositories are the author's own. [TODO: author to confirm that no other repository counted above contains third-party results]

Excluded material. An entire excluded lineage of earlier work, and every claim already marked [F], is excluded and never built on: 327 Phase 0 entries plus the five Phase 3 [F] verdicts, 332 in total (TOE_ATTEMPT.md l.25–30). Of the 332, 161 ledger entries belong to the excluded lineage; they were skipped by this skeleton and are not reproduced, quoted or summarised. The remaining excluded entries are falsified claims; where relevant they appear below with [F] and are not used as support.

Lean audit. Per TOE_ATTEMPT.md l.32–36: all eight Chair44 certificate files were re-checked by the Lean 4.35.0 kernel and use only standard axioms; 192 theorems whose names promise more than they prove were relabelled (23 Chair44, 38 QIS, 131 p_vs_np_repo); no Lean file defines P, NP or polynomial time; 139 ledger entries rest on hard-coded tests. The audit record is toe/inventory/lean_recheck.json with per-file logs (Lean 4.35.0, `lean <file>`, no Mathlib, no `sorryAx`, standard axioms only). These audit facts carry no label in the source and are not used as support for any labelled claim below. [TODO: labels for the audit facts: needs the ledger owner]

Refutations confirmed by the audit. [F] `S1-QIS-PNP` The QIS headline claim of an "unconditional proof that P $\neq$ NP" does not survive that repository's own held-out audit.

Evidence: TOE_ATTEMPT.md §8 l.401; CORE.md §5 l.84; ledger.json QIS-322 [F, excluded as falsified]

[F] `S1-QIS-PLANNER` The QIS "64–0 planner" record does not survive that repository's own audit.

[1] The ledger file of the source snapshot used here (toe/ledger.json, sha256 `46388050...`) lists 1288 claims and 332 excluded entries, and its `counts` field agrees. An older read-only copy (sha256 `a109117a...`) lists 1272 claims and 324 excluded entries while its `counts` field still says 1288/332 (discrepancies.md App. A). TOE_ATTEMPT.md §7 l.363 and §8 l.396–397 print the Phase 0 numbers 1243/327. This note uses 1288/332; the reconciliation is pending with the ledger owner.

Evidence: TOE_ATTEMPT.md §8 l.401; CORE.md §5 l.84; ledger.json QIS-182 [F, excluded as falsified]

3 The unified framework

3.1 The object

The definitions follow CORE.md §1. *Hull.* The tiles are chairs (a $2 \times 2 \times 2$ block of unit cubes minus one corner). A tile is a pose (G, T) with G a signed permutation matrix (its *frame*) and $T \in \mathbb{Z}^3$. A frame rule $d = (\pi_0, \ldots, \pi_7) \in S_3^8$ fixes the frames of the eight children; there are $6^8 = 1\,679\,616$ rules, of which $3^8 = 6561$ are proper. The hull Ω_d is the set of tilings of $\mathbb{Z}^3$ all of whose finite patches are legal; the level-L supertile T_L has 8^L tiles. The geometry does not depend on d; only the frames do. *Complex.* The panel multigraph $K(Y)$ of a tiling Y has the tiles as vertices and one edge (panel) per unit face shared by two tiles. Its 2-cells are the lattice-edge plaquettes and the two-cycles between pairs of tiles. Every tile has 24 panels, so $K(Y)$ is 24-regular. *Group and connection.* $G \in \{Q_8, 2I, SU(2)\}$, with $Q_8 = \{\pm 1, \pm i\sigma_k\}$ for frame rules, $SU(2)$ for the θ-family and the binary icosahedral group $2I$ as an intermediate step. A connection assigns $U(a \to b) \in G$ to each oriented panel, with $U(b \to a) = U(a \to b)^{-1}$. For a frame rule, $U_f(a \to b) = (\sigma \cdot M_a^T n_f)(\sigma \cdot M_b^T n_f) \in Q_8$ (the Pauli formula). The θ-family is $U_\theta(a \to b) = R_{u_a}(\theta) R_{u_b}(\theta)^{-1}$ with $R_u(\theta) = \cos\theta - i \sin\theta\, \sigma \cdot u$ and u the row of the frame along the panel's axis; at $\theta = \pi/2$ it is the Pauli link. *Holonomy, Laplacian, floor.* The holonomy $W(\gamma)$ is the ordered product of U along a closed walk; the connection is *frustrated* if some holonomy is non-trivial. With $(D_U \psi)_f = \psi_a - U_f(a \to b)\psi_b$, the covariant Laplacian is $H_U = D_U^\dagger D_U = 24 - A_U$, and the floor is $E_*(U) = \inf \sigma(H_U)$. On a pinned patch K one uses the compression of H_U to K, whose lowest eigenvalue is at least E_*. *Splitting.* Q_8 has five irreducible representations: the trivial one, three one-dimensional "shadows" χ_1, χ_2, χ_3, and the two-dimensional spinor. H_U splits accordingly into a graph Laplacian, three signed-graph Laplacians and one genuinely non-Abelian spinor block. *Unique-games reading.* (K, U) is a Q_8 unique game with one constraint $\psi_a = U(a \to b)\psi_b$ per panel; frustration means that no assignment satisfies every panel.

3.2 What is proved or certified about the core

(CORE.md §2; TOE_ATTEMPT.md §2 l.52–59.)[2]

[P] K1 On a connected finite graph with $SU(2)$ links: flat $\Leftrightarrow$ gauge-trivial $\Leftrightarrow \lambda_{\min}(H_U) = 0$. unverified

Evidence: CORE.md §2 l.38 (source: Chair44 X1)

[P] K2 E_* = inf over finite legal patches of $\lambda^D = \lim D_R$; it depends only on the set of legal environments, is order-reversing in it, and does not depend on frequencies. unverified

Evidence: CORE.md §2 l.39 (source: XII.A)

[P] K3 Exact IMS localisation identity; $\Lambda(\chi) \le E_* \le D_R$ with $D_R - \Lambda \le \varepsilon(\chi)$; rate R^{-2} on doubling or Euclidean graphs, and R^{-2} is optimal; property A alone gives no power law (lamplighter). unverified

Evidence: CORE.md §2 l.40 (source: XII.B, XII.C)

[P] K4 Every geometry-compatible frame field has ≥ 12 of the 21 centre-octant two-cycles non-trivial in every level-1 supertile, so the connection is never pure gauge. Lean kernel-checks only encoded finite predicates; this geometric statement is not itself a kernel-checked Lean theorem.

Evidence: CORE.md §2 l.41 (source: Phase VII-B P1)

[C] K5 $E_* > 0 \Leftrightarrow$ some T_k is not flat. All 1098 gauge classes of the 6561 proper rules have $E_* \ge 3.788$.

[2] TOE_ATTEMPT.md §5 l.328 calls K1–K15 "mathematical claims [P]"; CORE.md §2 labels K5 and K9 [C], K6 [P]/[C] and K10 [P]+[C]. The weaker CORE.md labels are used, and the compound ones are printed as [C] (Appendix C).

Evidence: CORE.md §2 l.42 (source: X3)

[C] K6 No lower bound on E_* follows from plaquette-class frequencies: there are two $\mathbb{Z}^3$ connections with the same Q_8 class on every plaquette and floors separated by ≥ 0.18. unverified

Evidence: CORE.md §2 l.43 (source: X4, X5, XII.P63); source label [P/C]

[P] K7 The non-spinor floors are functions of the plaquette class field; the X.5 separation lives in the spinor component. unverified

Evidence: CORE.md §2 l.44 (source: XI.H3)

[P] K8 Orbit Lemma: $\sigma_{dQ} = \rho_Q \circ \sigma_d$, ρ_Q is a global gauge transformation, $T_1(dQ) \cong T_1(d)$, and $E_*(dQ) = E_*(d)$ on the conjugation stabiliser; outside it the equivalence can fail from level 2. unverified

Evidence: CORE.md §2 l.45 (source: XII.OL)

[C] K9 $E_*(1387) \in [9.1104981, 9.3616608]$; $E_*(539311) \in [9.2083494, 9.3817818]$; the maximiser over all 6^8 rules is one of 1387, 539311, 539291.

Evidence: CORE.md §2 l.46 (source: XII.R20, XII.r24, XII.F4)

[C] K10 No function of level-one data determines E_* (1387 against 539288). unverified

Evidence: CORE.md §2 l.47 (source: XII.T31); source label [P+C]

[P] K11 For every Q_8 gauge section s of a patch K: $\lambda_{\min}(P_K H P_K) \leq B(s) = [\sum_e (2 - 2\,\mathrm{Re}\, d_e) + b(K)]/|K|$; the floor is bounded by the unique-games frustration energy of any section. unverified

Evidence: CORE.md §2 l.48 (source: XII.ug.P151)

[P] K12 The Clifford transports lie in Q_8; on vectors (the adjoint) the link group is the Klein group V_4; in two dimensions the Pauli-frame connections are $\mathbb{Z}_4$-valued (Abelian). unverified

Evidence: CORE.md §2 l.49 (source: CLIFFORD_Q8_PHASE2 Thms 2.2-2.3; X6)

[P] K13 No chirality: Wilson traces and plaquette classes are identical for a tiling and its mirror. unverified

Evidence: CORE.md §2 l.50 (source: CLIFFORD_Q8_PHASE2 Thm 2.4)

[P] K14 Diamagnetic bound: $\lambda_{\min}(H_U) \geq \lambda_{\min}(H_1)$ on every patch, for every unitary connection (Kato's inequality; standard). unverified

Evidence: CORE.md §2 l.51 (source: standard; prereg/G1_volume_scaling.json T1)

[P] K15 In infinite volume, independent ($\beta = 0$) links, and every finite-β Gibbs measure of a finite-group Wilson action, give floor 0 almost surely (finite energy plus Borel–Cantelli). unverified

Evidence: CORE.md §2 l.52 (source: Phase VII §8; prereg/G1_volume_scaling.json T3)

K4 is the only claim here that rests on Lean; its support is the kernel re-check logged in toe/inventory/lean_recheck.json together with an exhaustive enumeration (1 679 616 assignments) recorded in the ledger. The re-checked file is `certificates/Phase7Counting.lean`; the theorems that carry the encoded counting predicates are `octant_bound`, `total_bound` and `bound_attained` (Lean 4.35.0, standard axioms only, no `sorryAx`, as logged in lean_recheck.json). That `.lean` file is not part of the source snapshot used here, so the names are taken from the Chair44 Lean audit and the re-check log rather than from a fresh reading of the Lean statements. The audit records that Lean certifies only those encoded finite predicates (an encoded `trivial` test); the identification of that encoding with geometric two-cycle holonomy, and the step from per-octant bounds to "≥ 12 of 21", sit outside the kernel check. The geometric English of K4 is therefore not itself a Lean theorem statement. Theorems `forced_proper` and `forced_fails_improper` in the same file concern the proper-frame "nine forced" count, which is not K4. K5 combines an equivalence ("$E_* > 0$ iff some T_k is not flat") with a certified computation; the label [C] is the source's and covers the computation; the equivalence is not

separately labelled in the source. [TODO: split K5 into a [P] statement and a [C] computation once the author labels the equivalence] [TODO: cite: T. Kato, Israel J. Math. 13 (1972) 135, doi:10.1007/BF02760233 (Kato's inequality, K14) – not in verified .bib]

3.3 How the repositories fit

[P] I-TSEITIN Tseitin parity formulas are unsatisfiable iff the total charge is odd ($\mathbf{1}_V \in$ ker of the incidence over $\mathbb{F}_2$); this is the $\mathbb{Z}_2$ coboundary side of the core, in transposed form. unverified

Evidence: TOE_ATTEMPT.md §2 l.63; CORE.md §3 l.61

[P] I-ENERGY The continuous multilinear energy of a Tseitin instance has floor $E_{\min} \geq 1$. unverified

Evidence: CORE.md §3 l.62

[A] I-ENERGY-INST Read as an instance of the core, that floor is an analogy only: it is the floor of a relaxation energy, not of a covariant Laplacian.

Evidence: CORE.md §3 l.62

[P] I-XIII-SCOPE Scope statement: unique-games instances with $G = Q_8$ are the combinatorial side of the core; the external Unique Games Theorem is worst-case on finite instances and bounds no floor. (The external theorem itself is not checked here.) unverified

Evidence: TOE_ATTEMPT.md §2 l.64; CORE.md §3 l.60

[X] I-UG-EXT Third-party claims (Unique Games Theorem / Min-UnCut hardness; "Family 378: ETH is false") are recorded as external and unchecked.

Evidence: CORE.md §3 l.69

[E] I-QIS320 The Laplacian λ_2 of SAT clause graphs plateaus near 1.01 as N grows.

Evidence: CORE.md §3 l.64; ledger.json QIS-320

3

[A] I-QIS320-INST As an instance of the core this is an analogy only: a scalar (trivial-representation) graph Laplacian with no connection.

Evidence: CORE.md §3 l.64; §5 l.87

[F] I-QIS299 "Residual frustrated clauses carry a non-trivial 1-cycle" (a would-be holonomy statement for a $\mathbb{Z}_2/\mathbb{R}$ cochain) is falsified.

Evidence: CORE.md §3 l.65; ledger.json QIS-299

[C] I-GU-C4 The 4D chair C_4 has face counts $(80, 212, 210, 92, 15)$, $\chi = 1$; its boundary has $(80, 208, 192, 64)$, $\chi = 0$ (geometry only; no connection).

Evidence: CORE.md §3 l.67; ledger.json GU-001

[A] I-GU-A Geometric Unity: the fibre of $\mathrm{Met}(X)$ as an internal space is a structural analogy only (no shared object).

Evidence: CORE.md §3 l.68

[A] I-WOLFRAM Hypergraph rewriting (Wolfram Physics Project) is substitution-like rewriting of graphs with no group-valued connection on a fixed substitution hull: analogy only.

Evidence: CORE.md §3 l.70

[3] **Flag (I-QIS320):** ledger QIS-320 label E; source_label 'certified' (ledger did not upgrade)

Earlier dynamical-ensemble results. [F] `I-XIV-H1` Phase XIV (dynamical Q_8 Wilson ensemble) prediction H1: the hot-series $\langle\lambda_{\min}\rangle(\beta)$ is non-increasing within 3 SE — falsified.

Evidence: CORE.md §3 l.59; TOE_ATTEMPT.md §4 row 8 l.319; ledger.json CH14-XIV.H1

[F] `I-XIV-H2` Phase XIV prediction H2: $Y_3(1387)$ exceeds every sampled $\lambda_{\min}$ at every β — falsified.

Evidence: CORE.md §3 l.59; TOE_ATTEMPT.md §4 row 8 l.319; ledger.json CH14-XIV.H2

[E] `I-XIV-H3` Phase XIV H3 holds: the floor falls fastest where the inter-tile plaquette rises fastest (within one grid step).

Evidence: CORE.md §3 l.59; ledger.json CH14-XIV.H3

[F] `I-VI-017` The earlier (Phase VI) claim that random i.i.d. Q_8 links are gapped (floor ≈ 14.4) is withdrawn/falsified; G1 explains it as a pre-asymptotic plateau.

Evidence: TOE_ATTEMPT.md §3.1 l.115; ledger.json CH-VI-017

What does not fit. Random 3-SAT, solver and ML engineering, and the 6D→3D icosahedral modules have no group-valued connection (CORE.md §4). They are left out of the framework rather than forced into it: none of them defines a connection whose holonomy could be frustrated, so the floor E_* is not defined for them, and nothing proved about the core says anything about them.

Computed diffraction checks (outside the framework). Two computed-diffraction runs from the programme's materials work (quasicrystal-materials/diffraction/README.md, Turns 3 and 4) are recorded here so that they are not misread. Neither is a claim about E_*, the frustrated connection or Theorem F, and neither adds a labelled claim to Appendix A. Both are computed kinematic diffraction of generated point sets or occupation-domain models: not measured crystals, not structure refinements, and not observations. "[C on patch]" below means [C] for the finite generated patch only.

Chair44 at $L = 64$. On a generated FR30 level-6 patch (262 144 tiles; seven occupied boxes of side $L = 64$; frozen preregistration sha256 `10253440...`; table tables/chair44_L64.json), with decorations D1 (tile centroids) and D2 (cube centres weighted by frame) that are modelling choices [A], the dyadic controls keep $I_{64}/I_{32} \approx 1$ (H1 pass) [C on patch]. The denominator-64 test vector $q = (3/16, 5/32, 1/64)$ falls: $I_{64}/I_{32} = 0.0180$ for D1 and 0.0451 for D2 (stored values 0.01796463270465118 and 0.045059895785677664) [C on patch], so it is not a resolved Bragg peak on this patch. The preregistered non-dyadic control failed: at $(1/3, 0, 0)$ the D1 ratio is 0.2500777662631863, above the locked cut 0.25 (H3 fail; the cut was not moved). These decorated point sets index on the cubic module $\mathbb{Z}[1/2]^3$: they are limit-periodic cubic, not icosahedral, and they are not compared with any icosahedral powder list. A separate, unscored $q_z = 0$ image of the same patch (Figure 1; H1–H3 were not rerun) is descriptive only. For D1 the anti-diagonal $q_x + q_y = 1$ is still visible at $L = 64$ but fades [C on patch]: between the dyadic peaks its median intensity drops from about 4.21×10^{-6} ($L = 32$ boxes of the same patch) to about 2.31×10^{-7} ($L = 64$); at the same sixteen denominator-32 points on the line the median I_{64}/I_{32} is about 0.064 (10th–90th percentile 0.056–0.53); and at $L = 64$ the line still sits about 400× above off-diagonal denominator-64 control pixels (median about 5.59×10^{-10}).[4] A drop of about 1/16 per doubling of L is faster than both a Bragg peak (~ 1) and three-dimensional diffuse scattering ($\sim 1/8$), so on these patches the streak reads as a finite-box feature that fades, not a resolved Bragg or diffuse sheet [X]; that reading is not a statement about the infinite hull. For D2 the X-pattern along both diagonals is made of dyadic peaks of denominator at most 32 that stay

[4] Stored values (tables/chair44_L64_qz0_streak.json, sha256 `705d9d2d...`): D1 anti-diagonal median $4.209777756016015 \times 10^{-6}$ ($L = 32$) and $2.3131609108000826 \times 10^{-7}$ ($L = 64$); median ratio 0.06362729841650083 (p10 0.05580883569670223, p90 0.5255304654777284); control median $5.591859377559668 \times 10^{-10}$; D2 median ratio 0.8546743253992082; D2 denominator-64 anti-diagonal median $2.452103791329115 \times 10^{-7}$. The ratio of the D1 line to the control at $L = 64$ is 413.67.

put (median $I_{64}/I_{32} \approx 0.855$), and the denominator-64 anti-diagonal median is about 2.45×10^{-7} [C on patch]. Finite-patch numbers are not a proof that the infinite hull is limit-periodic; that ledger statement (CH-V-020) remains a conjecture [X], and unique ergodicity of the framed-chair hull remains an unverified input (§4.2).

Icosahedral occupation-domain models. Separately, on the icosahedral cut-and-project side that the previous paragraph leaves outside the framework, multi-window occupation-domain decorations were scored against two published powder lists [E] (frozen preregistration sha256 `c8e0a3c0...`; table tables/multiwindow_pf.json, sha256 `d6668b04...`). This is independent work of the programme: no Chair44 file was used, CH-V-020 was not touched, and none of it is a Chair44, E_* or Theorem F statement. The model inputs are assumptions [A]: occupation-domain radii transferred from an isostructural Tsai-type model, volume-equivalent spheres, published truncations and shifts omitted, and an F-type node split that tests splitting rather than a new overall scale. P-type against the $Zn_{75}Ni_{10}Sc_{15}$ list of Maezawa, Kashimoto and Ishimasa [16] (Table 1) [E]: the chemistry-aware Spearman rank correlation is $\rho = 0.338$ (stored 0.3378353860555397; geometric 0.336) against 0.553 for the same-run single-window control, so the preregistered gain of +0.05 was not reached (H1 fail, $\Delta = -0.215$). The two lines that the single-window model predicted but the list does not contain, $(N, M) = (70, 113)$ and $(6, 5)$, drop to $I = 2.27$ and 3.28 on a scale with the strongest matched line at 100, below the locked cut 20 (H2 pass) [C]. F-type against the icosahedrite $Al_{63}Cu_{24}Fe_{13}$ list of Bindi et al. [3] (Table 1) [E]: $\rho = 0.574$ against 0.603 for the same-run control (H3 fail, $\Delta = -0.029$), and the unobserved $(60, 96)$ line is still predicted at $I = 66.3$, above the locked cut 20 (H4 fail). The readings recorded with that work stay [X]: the vertex-decorated single-window model is not a structure model for Tsai-type phases, and the single-window F-type model predicts a strong line that is not observed.

3.4 Three extensions used in Phase 3

(1) Ensembles of connections from the Wilson measure $\exp(-\beta \sum_p (1 - \frac{1}{2}\mathrm{Re}\,\mathrm{Tr}\,U_p))$ on the rigid-tile complex, $G \in \{Q_8, 2I, SU(2)\}$ [29] (a standard pointer for the Wilson action; the action's normalisation and its use with the finite groups Q_8 and $2I$ are not checked from the abstract, and the journal body was not re-opened for this draft); (2) the $SU(2)$ deformation U_θ followed along $\theta = \kappa a$, $a = 2^{-L}$; (3) a time direction, $T_L \times \mathbb{Z}_{N_t}$ with its transfer matrix (TOE_ATTEMPT.md l.67–70).

4 Results, gap by gap

Each gap followed the same procedure: a frozen preregistration committed before any value it predicts was computed; implementation and run; scoring with the frozen rules, no refitting (TOE_ATTEMPT.md l.74–79). The source was read from a file snapshot, not a git checkout, so the freeze order cannot yet be checked against commit history; Appendix D lists the file hashes, seeds and run parameters that are known. [TODO: prereg commit hashes and timestamps (G1, G2, G3, G5) and the toe repository commit: needed from the author]

4.1 Gap 1: volume scaling of the matter floor

Model. Independent Haar links ($\beta = 0$) on pinned supertiles $T_1 \dots T_5$ (8 to 32 768 tiles), groups Q_8, $2I$, $SU(2)$, 2780 samples (l.83–87).

Theorems used. [P] `G1-T1` T1: the diamagnetic bound and compression give $\lambda_1(T_L) \le \lambda(T_L) \le$ min over the eight child supertiles. unverified

Evidence: TOE_ATTEMPT.md §3.1 l.90

[P] G1-T2 T2: the children are i.i.d. copies of T_{L-1}, so $P(\lambda(T_L) > x) \le P(\lambda(T_{L-1}) > x)^8$. unverified

Evidence: TOE_ATTEMPT.md §3.1 l.91

[P] G1-T3 T3: for every $\beta \in [0, \infty)$ and every infinite-volume Gibbs measure of a finite-group Wilson action on the hull, $\inf \sigma(H_U) = 0$ almost surely (standard argument; not claimed new). unverified

Evidence: TOE_ATTEMPT.md §3.1 l.92-94; ledger.json TOE-G1-T3

[5]

[X] G1-RARE Rare-region scale: a trivial 1-star has probability $8^{-24} \approx 2 \cdot 10^{-22}$ at $\beta = 0$, so the regions that drive the floor to 0 first appear at about 10^{22} tiles, far beyond T_5. [label pending author ruling on 'P-level estimate']

Evidence: TOE_ATTEMPT.md §3.1 l.95; source label [P-level estimate]

[6]

Proof of T3 (from TOE_ATTEMPT.md l.93; hand proof). Let G be finite and n_p the number of plaquettes per link. Given everything outside a ball B, the conditional probability that all links of B are trivial is at least $(e^{-2\beta n_p}/|G|)^{|B|} =: \delta_r > 0$, uniformly in the outside configuration. Over infinitely many disjoint balls of radius r, successive conditioning gives $P(\text{no trivial ball among the first } n) \le (1 - \delta_r)^n \to 0$, so a trivial ball of radius r exists almost surely, for every r. The Dirichlet ground state of a trivial ball of radius r is a test vector for H_U with Rayleigh quotient tending to 0 as $r \to \infty$. Hence $\inf \sigma(H_U) = 0$ almost surely. □ This write-up is the author's and is not claimed new; it extends an earlier $\beta = 0$ argument of the author to every β. Nakamura records $\inf \sigma = 0$ almost surely in a different model (i.i.d. $U(1)$ fields on $\mathbb{Z}^2$) [18]; that paper is not the source of the finite-energy and Borel–Cantelli argument above, and it does not treat Gibbs measures at $\beta > 0$, non-abelian finite groups, or a substitution hull.

Results (simulated). [E] G1-FLOORS Q_8 mean floors on $T_1 \dots T_5$ ($\beta = 0$, 2780 samples): 18.511, 16.024, 15.036, 14.615, 14.367 ($\pm$0.027, 0.015, 0.009, 0.008, 0.014).

Evidence: TOE_ATTEMPT.md §3.1 l.98; results/g1_score.json

[E] G1-TRIVIAL The trivial-connection floor falls from 11.21 to 0.139 over $T_1 \dots T_5$.

Evidence: TOE_ATTEMPT.md §3.1 l.99

[E] TOE-G1-T1 Every $\beta = 0$ sample obeys T1 (0 violations).

Evidence: TOE_ATTEMPT.md §3.1 l.100; ledger.json TOE-G1-T1 [C]

[7]

[E] TOE-G1-T2 T2 holds at every level: mean $\beta = 0$ floor of $T_L \le E[\text{min of 8 i.i.d. level-}(L-1)\text{ floors}] + 3\,\text{SE}$, $L = 2..5$.

Evidence: TOE_ATTEMPT.md §3.1 l.100; ledger.json TOE-G1-T2

Scores (frozen rules). [F] TOE-G1-H1 H1 (preregistered): the ground state sits (weight $\ge \frac{1}{2}$) on the weakest child in $\ge 2/3$ of samples. Outcome: fractions 0.345, 0.450, 0.650 ($L = 3, 4, 5$).

Evidence: TOE_ATTEMPT.md §3.1 l.106; ledger.json TOE-G1-H1

[F] TOE-G1-H2 H2 (preregistered): min-stable median law $\hat{m} = 0.0830$-quantile of level $L-1$. Outcome: medians below the band at $L = 3, 4$, inside at $L = 5$.

[5] **Flag (G1-T3):** ledger evidence cites stale 'TOE_ATTEMPT.md s4.1' (now §3.1)

[6] **Flag (G1-RARE):** non-standard source label 'P-level estimate'; printed [X] pending ruling

[7] **Flag (TOE-G1-T1):** LABEL MISMATCH: TOE_ATTEMPT lists this under 'Results [E, simulated]', ledger says [C]; weaker [E] used (no upgrade); needs ledger reconciliation

Evidence: TOE_ATTEMPT.md §3.1 l.107; ledger.json TOE-G1-H2

[E] TOE-G1-H3a H3a holds: $2I$ and $SU(2)$ floors agree within 3σ ($\Delta = +0.016, -0.006, +0.024$).
Evidence: TOE_ATTEMPT.md §3.1 l.108; ledger.json TOE-G1-H3a

[F] TOE-G1-H3b H3b (preregistered): Q_8 significantly below $SU(2)$. Outcome: right sign ($-0.021, -0.014, -0.017$) but never beyond 3σ.
Evidence: TOE_ATTEMPT.md §3.1 l.109; ledger.json TOE-G1-H3b

[E] TOE-G1-H4 H4 holds (weak test, recorded as such): medians (18.53 ... 14.39) lie above the certified $E_*(1387) \geq 9.1104981$.
Evidence: TOE_ATTEMPT.md §3.1 l.110; ledger.json TOE-G1-H4

Reading. A descriptive, non-preregistered geometric extrapolation of the decrements (2.49, 0.99, 0.42, 0.25) would suggest an infinite-volume gap near 14; T3 shows this is false, and the plateau is pre-asymptotic (l.113–115). The extrapolation is not a claim of this note and is not labelled.

[P] SC-8 Scorecard row 8 (programme-internal), settled negatively: in infinite volume the matter floor is 0 at every β for finite groups (G1 T3), so it is not an infinite-volume order parameter of any finite-group Wilson ensemble. unverified
Evidence: TOE_ATTEMPT.md §3.1 l.116; §4 row 8 l.319
[8]

4.2 Gap 2: a scaling region and a continuum limit

Part B: the θ-trajectory of the frame connection

Model. U_θ, rules FR30 and 1387, levels $T_1 \ldots T_6$ (up to 262 144 tiles); along $\theta_L = \pi 2^{-L}$ the gauge part of the floor in physical units is $G_L = 4^L(\lambda(\theta_L) - \lambda_1) \to \pi^2 c_\infty$ provided c_L converges (l.126–130).

[E] G2B-DERIV Derivation: $\lambda(T_L, \theta) = \lambda_1(T_L) + c_L\theta^2 + O(\theta^3)$, with c_L = dia − para given exactly by second-order perturbation theory about the Dirichlet ground state. [label pending author ruling on 'P-level']
Evidence: TOE_ATTEMPT.md §3.2 l.129; source label [P-level]
[9] (G2B-DERIV is the Phase 3 form of the expansion; its proved form is Theorem A below, TOE-PROOF-A.)

[E] G2B-CL Floating-point c_L: FR30 8.704, 11.324, 12.674, 13.046, 13.243, 13.318; 1387 9.850, 12.458, 13.685, 14.039, 14.196, 14.253; $G_6/(\pi^2 c_6) = 0.9992$ for both. The frozen B2 geometric extrapolation from (c_4, c_5, c_6) gives $c_\infty \approx 13.364$ (FR30), 14.285 (1387); from (c_3, c_4, c_5) the same formula gives 13.463 and 14.322 (four-decimal inputs), so the extrapolated value moves by 0.10 and 0.04 with the fitting range. The geometric model $c_L = c_\infty - Aq^L$ has three parameters and is fitted to three points exactly, so no goodness-of-fit statistic exists; the range sensitivity is the only check. Both lie inside the certified brackets of Corollary E.
Evidence: TOE_ATTEMPT.md §3.2 l.124,132-135; results/g2_partB.json

All c_L are deterministic floating-point solves, so they carry no statistical error bars: the resolvent equations are solved by conjugate gradients with relative tolerance 10^{-12} (src/pert.py), and the recorded maximum relative residual is about 10^{-12} or below (results/g2_partB.json). The B2 extrapolation formula $c_\infty = c_6 + (c_6 - c_5)r/(1-r)$, $r = (c_6 - c_5)/(c_5 - c_4)$, was fixed in the preregistration (prereg/G2_scaling.json); its sensitivity to the fitting range is given in the claim.

[8] **Flag (SC-8):** README.md says 'One scorecard row is [P]' but row 8 is '[P] negative' (discrepancies.md §2.7)
[9] **Flag (G2B-DERIV):** non-standard source label 'P-level'; printed [E] pending ruling; the proved version is TOE-PROOF-A

[E] TOE-G2-B0 B0 holds (a check of the code, not physics): finite-θ Taylor ratios 0.998–0.999; an independent finite-difference test agrees to 10^{-3} (floating point, no error control).

Evidence: TOE_ATTEMPT.md §3.2 l.124,139; ledger.json TOE-G2-B0; source label [C]

[E] TOE-G2-B1 B1 holds: geometric convergence, difference ratios FR30 0.528, 0.380; 1387 0.444, 0.359 (≤ 0.75).

Evidence: TOE_ATTEMPT.md §3.2 l.140; ledger.json TOE-G2-B1

[E] TOE-G2-B2 B2 holds: $c_6 > 0$ and $c_\infty > 0$ (13.36, 14.28).

Evidence: TOE_ATTEMPT.md §3.2 l.141; ledger.json TOE-G2-B2

[E] TOE-G2-B3 B3 holds: $G_6/(\pi^2 c_6)$ within 10 % of 1 (0.9992, 0.9992; from 0.17 at $L = 1$ through 0.94, 0.987, 0.997). Uniformity of the θ^3 remainder is not proved, so B3 remains [E].

Evidence: TOE_ATTEMPT.md §3.2 l.142,153; PROOF.md §0 not-proved item 1; ledger.json TOE-G2-B3

[F] TOE-G2-B4 B4 (preregistered): the 20 lowest levels shift uniformly (± 20 % of c_5) at $L = 5$. Outcome: entries 18 and 20 are 0.925 and 0.559 (FR30), 0.929 and 0.590 (1387).

Evidence: TOE_ATTEMPT.md §3.2 l.143; ledger.json TOE-G2-B4

[X] G2B-LIMIT Conjecture: along $\theta = \kappa a$ the frame connection has a continuum limit (the computed G_L, $L \leq 6$, are consistent with this; existence is not shown).

Evidence: TOE_ATTEMPT.md §3.2 l.146; §7 l.371; source label [E]

[X] G2B-MASS Conjecture: in that limit the frustration survives only as a finite mass term, $\Lambda_{\text{phys}} =$ (Dirichlet Laplacian of the chair region) $+\kappa^2 c_\infty$, with c_∞ the limit of the derived (not fitted) coefficients c_L.

Evidence: TOE_ATTEMPT.md §3.2 l.147; §4 row 7 l.318; §7 l.372; source label [E]

[E] G2B-POSTHOC Post-hoc B4 diagnostic (not preregistered; does not change the frozen [F]): rerun with $k = 28$ puts all 20 index-wise shifts in $[0.985, 0.997]c_5$ (FR30), $[0.984, 0.997]c_5$ (1387); the uniform-mass-shift prediction is plausible but not established.

Evidence: TOE_ATTEMPT.md §3.2 l.159-164; src/posthoc_b4.py; source label [E, post hoc]

Reading [X]. For the linearised θ-family only, and if the conjectures G2B-LIMIT and G2B-MASS hold, no non-Abelian field survives the limit: the rapidly oscillating frame field homogenises and only a mass term remains (TOE_ATTEMPT.md l.156–157). This reading is a conjecture; it is unlabelled in the source and printed here as [X]. An alternative reading is not excluded by the data: the six levels may still be pre-asymptotic (the B1 ratios fall from 0.53 to 0.38 for FR30), and the second-order coefficient says nothing about higher orders in θ, where interacting terms could survive.

Proved afterwards: the $\theta \to 0$ limit (PROOF.md)

PROOF.md states that its proof "is written and checked here; it is not machine-checked and not peer-reviewed" (PROOF.md §0 item 4); all its [P] items therefore carry unverified.

[P] TOE-PROOF-A Theorem A: $\lambda(T_L, \theta) = \lambda_1(T_L) + c_L \theta^2 + O(\theta^3)$, with $c_L = \min_G \sum_e \varphi_L(a)\varphi_L(b)\,|w_e + G_b - G_a|^2$. unverified

Evidence: PROOF.md §0 l.10, §2; ledger.json TOE-PROOF-A

[P] TOE-PROOF-B Theorem B: the twisted free densities μ_ℓ increase; the corrected Dirichlet densities $\tilde{\nu}_\ell = \nu_\ell^0 + 48 \cdot 2^{-\ell}$ decrease; $\mu_\ell \leq \nu_\ell^0$. unverified

Evidence: PROOF.md §0 l.11, §3; ledger.json TOE-PROOF-B

[P] TOE-PROOF-C Theorem C: for every frame rule and every $\ell \le L$, with explicit $\varepsilon_{\ell,L} \to 0$: $\mu_\ell(1-\varepsilon_{\ell,L}) \le c_L \le \tilde{\nu}_\ell(1+\varepsilon_{\ell,L})$. unverified

Evidence: PROOF.md §0 l.12, §4; ledger.json TOE-PROOF-C

[C] TOE-PROOF-D D: for every one of the $6^8 = 1\,679\,616$ frame rules, $\liminf_L c_L \ge \mu_1 \ge 3527/1012 > 3.485$ (attained at rule 128462). unverified

Evidence: PROOF.md §0 l.13, §5; ledger.json TOE-PROOF-D [C]; results/proof_positivity.json; source label [P+C]

[C] TOE-PROOF-E-FR30 E (FR30): $13.232574 \le \liminf c_L \le \limsup c_L \le 15.465590$ (level $\ell = 6$). The two numbers are certified computations; that they bound $\liminf c_L$ and $\limsup c_L$ (and, via Theorem F, ρ(FR30)) is conditional on Theorems B, C and F, which are hand proofs.

Evidence: PROOF.md §0 l.14, §5 l.184; ledger.json TOE-PROOF-E-FR30

[10]

[C] TOE-PROOF-E-1387 E (1387): $14.161980 \le \liminf c_L \le \limsup c_L \le 16.390966$ (level $\ell = 6$). Conditional on Theorems B, C and F as for FR30.

Evidence: PROOF.md §0 l.14, §5 l.185; ledger.json TOE-PROOF-E-1387

[C] PROOF-E-TABLE Certified level densities $\mu_\ell \ge$, $\nu_\ell^0 \le$ for FR30 and 1387, $\ell = 1..6$, printed in Table 1 rounded outward from the stored full-precision bounds.

Evidence: PROOF.md §5 l.173-182; results/proof_bounds.json

[11]

[C] TOE-PROOF-prim The frame substitution is primitive for 1 666 486 of the 6^8 rules (FR30, 1387: $k = 2$, $|H| = 24$); it fails for 13 130 rules, among them x539311 and x539291 ($|H| = 48$, period-2 walk).

Evidence: PROOF.md §6 l.194-195; ledger.json TOE-PROOF-prim; src/proof_primitivity.py

[P] TOE-PROOF-F Theorem F: for every rule whose frame substitution is primitive, c_L converges and $\lim c_L = \rho(d) := 7\,\min$ over stationary closed tile-compatible fields ξ of $E_\mu[\sum_k |w_k + \xi_k|^2]$. This is an application of standard ergodic homogenisation of discrete operators with stationary coefficients (Section 7.1), using unique ergodicity of the hull [25, 26], a multiparameter ergodic theorem [28] and Kolmogorov–Riesz compactness [8]. Unique ergodicity of this hull and the cube-versus-ball form of the ergodic input remain unverified (Section 4.2); they are not treated as newly checked [P]. unverified

Evidence: PROOF.md §0 l.15, §6; ledger.json TOE-PROOF-F

[P] PROOF-L7 Lemma 7 (sublinearity): for $\zeta \in \mathcal{C}_0$, $F_R(Y) := R^{-5} \sum_{q \in Q_R} |G_Y(q) - \langle G_Y \rangle_{Q_R}|^2 \to 0$, μ-a.s. and in $L^1(\mu)$, where $Q_R = \{0, \dots, R-1\}^3$ or any translate of it by $O(R)$. unverified

Evidence: PROOF.md §6 l.221-225

How the certified numbers depend on the hand proofs. The numbers μ_ℓ and ν_ℓ^0 are certified computations [C]. The certificate is a floating-point computation with explicitly bounded rounding error (PROOF.md §5 "Rounding" and Lemma 6), not interval arithmetic, so the [C] status itself rests on two further hand-proved estimates. That the numbers bracket $\liminf c_L$ and $\limsup c_L$ uses Theorems B and C; that they bracket $\rho(d) = \lim c_L$ uses Theorem F. All three are hand proofs [P, unverified]. The brackets are therefore conditional statements.

[10] **Flag (TOE-PROOF-E-FR30):** ledger prints upper end 15.465589, PROOF.md 15.465590; stored value 15.465588903524477, so both are outward-safe. PROOF.md value kept

[11] **Flag (PROOF-E-TABLE):** v0.0 printed PROOF.md's values; 10 cells were nearest-rounded (discrepancies.md §1). v0.1 prints outward values from results/proof_bounds.json; PROOF.md values are given in table footnotes

Scope of the cited theorems. PROOF.md §6.1 cites Solomyak [25, 26] for unique ergodicity, identifies the substitution matrix with the frame-walk matrix, and passes from $\mathbb{R}^3$ to $\mathbb{Z}^3$ through the suspension. Both steps are stated briefly in the source. Unique ergodicity of this framed-chair hull remains an unverified input. Solomyak's 1999 correction removes the implication from matrix primitivity to repetitivity; the original 1997 theorem's full hypotheses have not been checked from its body here. The later Lee–Moody–Solomyak paper [14], Theorem 4.1, proves unique ergodicity for FLC fixed points of primitive geometric tile substitutions without a separate repetitivity assumption, but the identification of our frame walk with such a substitution and the passage to the discrete $\mathbb{Z}^3$ suspension base remain unchecked. This [S] step is not treated as newly checked [P], and citing [14] does not check it. A novelty-note scope check (theorem-F-novelty.md §5.3) finds both the substitution-matrix identification and the $\mathbb{R}^3 \to \mathbb{Z}^3$ suspension step plausible but stated only briefly in the source; a later scope check (solomyak-S-scope-check-2026-11-07.md) could not obtain the 1997 or 1999 paper bodies. Wiener's paper [28] is cited in PROOF.md for the multiparameter pointwise ergodic theorem with cube averages; Wiener states his theorem for averages over balls of a continuous $\mathbb{R}^n$-action, while Lemma 7 uses cubes Q_R in $\mathbb{Z}^3$. This cube-versus-ball mismatch is a documented gap, not a newly verified ergodic input. Suggested cube-average theorems (N. Dunford, Acta Sci. Math. Szeged 14 (1951) 1; A. Zygmund, Fund. Math. 38 (1951) 1) are not in the verified bibliography and are not cited here.

A provisional seventh level. A level-$\ell = 7$ certificate run has been queued and run but its results are not committed to the source (compute-queue/queue.jsonl, job `toe-l7-cert`; result files on the author's machine). Its notes give $\mu_7 \geq 13.291755$, $\tilde{\nu}_7 \leq 14.480830$ (FR30) and $\mu_7 \geq 14.219208$, $\tilde{\nu}_7 \leq 15.407377$ (1387), with $\tilde{\nu}_7 = \nu_7^0 + 48 \cdot 2^{-7}$. Whether these values were rounded outward has not been checked; padding each end by 10^{-6} gives $[13.291754, 14.480831]$ and $[14.219207, 15.407378]$, both inside the $\ell = 6$ brackets and both containing c_6 and the extrapolated c_∞. Label: [C] [PROVISIONAL: uncommitted run, rounding direction unverified]. These numbers are not used anywhere else in this note; the headline brackets remain the $\ell = 6$ ones.

Table 1: Certified level densities [C], rounded outward to six decimals from the stored full-precision bounds (toe/results/proof_bounds.json, sha256 `b6752b18...`): lower bounds rounded down, upper bounds rounded up. Ten cells differ from the values printed in PROOF.md §5 (l.173–182); they are marked [a] and the printed value is given below.

ℓ	$\mu_\ell \geq$ (FR30)	$\nu_\ell^0 \leq$ (FR30)	$\mu_\ell \geq$ (1387)	$\nu_\ell^0 \leq$ (1387)
1	6.780961[a]	16.500001[a]	7.197628	17.250001[a]
2	10.596162	21.194785	11.401115	22.158938[a]
3	12.203877[a]	20.507430	13.276907[a]	21.414960
4	12.841639	17.829293	13.812599[a]	18.735685
5	13.109415	15.858983[a]	14.046759[a]	16.779928
6	13.232574	14.715589	14.161980	15.640966[a]

[a] Value printed in PROOF.md: FR30 $\mu_1 = 6.780962$, $\nu_1^0 = 16.500000$, $\mu_3 = 12.203878$, $\nu_5^0 = 15.858982$; 1387 $\nu_1^0 = 17.250000$, $\nu_2^0 = 22.158937$, $\mu_3 = 13.276908$, $\mu_4 = 13.812600$, $\mu_5 = 14.046760$, $\nu_6^0 = 15.640965$. PROOF.md says "All values are rounded outward", but these ten were rounded to nearest and lie on the unsafe side of the stored bounds by at most $5 \cdot 10^{-7}$ (discrepancies.md §1). The source file is unchanged; correcting it is the author's decision.

[b] Headline brackets: FR30 [13.232574, 15.465590], 1387 [14.161980, 16.390966] (PROOF.md l.184–185). Both are on the safe side. For FR30 the stored upper end is $\nu_6^0 + 48/64 = 15.465588903524477$; its tight outward value is 15.465589, which the ledger prints, and PROOF.md prints the valid but looser 15.465590. This note keeps 15.465590.

Not proved in PROOF.md. The scaling trajectory (B3 stays [E]); the 13 130 imprimitive rules (including x539311, x539291), for which only D holds; the uniform spectral shift B4; machine checking or independent review (PROOF.md §0 l.21–25).

Part A: $Q_8 \to 2I \to SU(2)$ Wilson ensembles

Model. Rigid-tile complex on T_2 (576 links) and T_3 (5376 links); $\beta \in \{0.5, \ldots, 64\}$; two chains, 500 thermalisation sweeps and 200 measurements per point (l.169–172). D2 (an angular-resolution estimate, unlabelled in the source and not used as support) gives $\beta_\times(2I) = 12.4$, $\beta_\times(Q_8) = 2.0$ (l.178). [TODO: label for D2: needs the author; it is an estimate, so no stronger than [X]]

[E] `TOE-G2-D1` D1 (weak coupling): $\beta\langle\lambda - \lambda_1\rangle \to C_{\text{pert}} = 3[\sum_e K^+_{ee}\varphi_0(a)\varphi_0(b) - \text{tr}(K^+J)]$, by second-order perturbation theory with the Gaussian Wilson measure; $C_{\text{pert}} = 11.4856$ (T_2), 13.6306 (T_3). The derivation is at the level of formal perturbation theory.

Evidence: TOE_ATTEMPT.md §3.2 l.175-177,203; ledger.json TOE-G2-D1; src/pert.py; source label [P] [12]

Table 2: G2 Part A [E]: $R = \beta(\langle\lambda\rangle - \lambda_1)/C_{\text{pert}}$ with one standard error, from toe/results/g2_score.json (Part A table). $C_{\text{pert}} = 11.4856$ (T_2), 13.6306 (T_3). Per point: two chains, 500 thermalisation sweeps, 200 measurements every 5 sweeps; the hot chain is scored at $\beta \le 2$, the cold chain at $\beta \ge 4$. Seed 20261031 (prereg/G2_scaling.json).

β	$SU(2)$, T_2	$SU(2)$, T_3	$2I$, T_3	Q_8, T_3
0.5	0.4498(8)	0.4505(5)	0.4504(4)	0.4472(5)
1	0.8010(21)	0.8067(9)	0.8072(10)	0.7775(11)
2	1.1614(38)	1.1961(19)	1.1993(19)	0.0215(4)
4	1.1388(54)	1.1825(24)	1.1786(25)	0.0000
8	1.0633(58)	1.0813(24)	0.8753(26)	0.0000
16	1.0331(47)	1.0381(23)	0.0010(1)	0.0000
32	1.0249(42)	1.0167(21)	0.0000	0.0000
64	1.0017(43)	1.0103(21)	0.0000	0.0000

Entries 0.0000: frozen ensemble, floor equal to λ_1 to four decimals (standard error below 10^{-4}).

[E] `TOE-G2-W1` W1 holds: $R_{SU2}(64) \in [0.85, 1.15]$, $R_{SU2}(32) \in [0.75, 1.25]$ (1.002/1.025 on T_2; 1.010/1.017 on T_3).

Evidence: TOE_ATTEMPT.md §3.2 l.195; ledger.json TOE-G2-W1

[E] `TOE-G2-W2` W2 holds: $|R_{SU2} - 1|$ non-increasing over $\beta = 8 \ldots 64$.

Evidence: TOE_ATTEMPT.md §3.2 l.196; ledger.json TOE-G2-W2

[E] `TOE-G2-W3a` W3a holds: $2I = SU(2)$ at $\beta \le 4$.

Evidence: TOE_ATTEMPT.md §3.2 l.197; ledger.json TOE-G2-W3a

[E] `TOE-G2-W3b` W3b holds: $2I$ frozen ($R_{2I} < R_{SU2}/2$) at $\beta = 32, 64$.

Evidence: TOE_ATTEMPT.md §3.2 l.198; ledger.json TOE-G2-W3b

[F] `TOE-G2-W3c` W3c (preregistered): $Q_8 = SU(2)$ at $\beta \le 1$ and frozen at $\beta \ge 4$. Outcome: frozen from $\beta = 4$ as predicted, but already different at $\beta = 0.5, 1$ (T_3: 13.862 vs 13.950; 12.267 vs 12.666).

Evidence: TOE_ATTEMPT.md §3.2 l.199; ledger.json TOE-G2-W3c

[E] `TOE-G2-W4` W4 holds: $2I$ within 25 % of $SU(2)$ at $\beta = 8$ (ratios 0.85, 0.81).

Evidence: TOE_ATTEMPT.md §3.2 l.200; ledger.json TOE-G2-W4

[12] **Flag (TOE-G2-D1):** ledger [P]; TOE text: 'at the level of formal perturbation theory'; printed [E] (Appendix C)

[E] G2A-SELFENERGY In the dynamical $SU(2)$ theory the gauge-induced floor shift approaches C_{pert}/β monotonically and agrees with it to 1 % at $\beta = 64$ on both levels (the floor is a mass renormalisation).

Evidence: TOE_ATTEMPT.md §3.2 l.203-205; §7 l.369

Reading. $2I$ has a narrow perturbative window ($\beta \approx 2$–8) and then freezes between $\beta = 8$ and 16, bracketing the estimate 12.4; Q_8 deviates earlier than estimated (W3c [F]) (l.206–208). Table 2 shows the pattern: $SU(2)$ approaches $R = 1$ from above for $\beta \geq 4$; $2I$ follows $SU(2)$ up to $\beta = 4$, drops to 0.875 at $\beta = 8$ and is frozen from $\beta = 16$; Q_8 already departs at $\beta = 1$ (0.7775 ± 0.0011 against 0.8067 ± 0.0009) and is frozen from $\beta = 2$. In raw floors the $\beta \leq 1$ departure of Q_8 is 13.862 ± 0.014 against 13.950 ± 0.014 at $\beta = 0.5$, and 12.267 ± 0.016 against 12.666 ± 0.013 at $\beta = 1$ (T_3).

4.3 Gap 3: a time direction

Model. Wilson theory on $T_2 \times \mathbb{Z}_8$ with rigid tiles, $\beta \in \{0.25, 0.5, 1\}$, 20 000 heat-bath measurements per run [10] (l.212–215). Two-link plaquettes (483 per slice on T_2) give $r(1) = u_G(\beta)^2$ at leading order and $m = -2 \ln u_G$ (l.222–225).

[P] TOE-G3-R0 R0: in temporal gauge the weight factorises as $M^{1/2} K M^{1/2}$ with positive-definite kernel $K(U, U') = \prod_e f(U_e U_e'^{-1})$, $f(g) = e^{\beta \operatorname{Re} g}$; the transfer matrix is self-adjoint and positive on any spatial 2-complex, so $H = -\ln T$ generates unitary e^{-iHt} on the gauge-invariant space (Lüscher 1977; Osterwalder–Seiler 1978; not chair-specific). unverified

Evidence: TOE_ATTEMPT.md §3.3 l.217-220; ledger.json TOE-G3-R0

Lüscher constructs a self-adjoint, strictly positive, gauge-invariant transfer matrix for $SU(N)$ on a four-dimensional hypercubic lattice [15]; that mechanism does not cover Q_8, $2I$, or an arbitrary spatial 2-complex. Osterwalder–Seiler is cited as a standard pointer for a positive self-adjoint transfer matrix [21] (abstract only; the proof, gauge choice, and group class were not seen). R0's extension to the present setting remains [P, unverified].

[E] TOE-G3-R1 R1 holds: $C(\tau) \geq 0$ and log-convex in every run, within 3σ.

Evidence: TOE_ATTEMPT.md §3.3 l.231; ledger.json TOE-G3-R1

[E] TOE-G3-R2 R2 holds: $r(1)/u^2 = 0.981, 1.011, 1.018$ ($\beta = 0.25$); $0.996, 1.073, 0.995$ ($\beta = 0.5$) for $2I$, Q_8, $SU(2)$.

Evidence: TOE_ATTEMPT.md §3.3 l.232; ledger.json TOE-G3-R2

[E] TOE-G3-R3a R3a holds: Q_8 above $SU(2)$ at $\beta = 1$ ($r(1) = 0.1184 \pm 0.0008$ vs 0.0534 ± 0.0002).

Evidence: TOE_ATTEMPT.md §3.3 l.233; ledger.json TOE-G3-R3a

[E] TOE-G3-R3b R3b holds: $2I = SU(2)$ within 3σ at every β.

Evidence: TOE_ATTEMPT.md §3.3 l.234; ledger.json TOE-G3-R3b

[E] TOE-G3-R3c R3c holds: m_{eff} within $\times 1.5$ of $-2 \ln u$ at $\beta = 1$ (Q_8 2.13 vs 2.70; $2I$ 2.92 vs 2.85; $SU(2)$ 2.93 vs 2.85).

Evidence: TOE_ATTEMPT.md §3.3 l.235; ledger.json TOE-G3-R3c

[E] G3-SPECTRUM The strong-coupling spectrum is the one derived, including the chair-specific mass $-2 \ln u_G$ from two-link plaquettes.

Evidence: TOE_ATTEMPT.md §3.3 l.238; §7 l.379

[P] SC-6 Scorecard row 6: quantum mechanics with unitary time evolution holds at the lattice level only (standard construction). unverified

Evidence: TOE_ATTEMPT.md §4 row 6 l.317; §4 l.321

Uncertainties for G3. Each G3 run has 20 000 measurements (results/g3_score.json). One standard error of $r(1)/u^2$, computed as $\sigma_{r(1)}/u^2$ from that file: $\beta = 0.25$: 0.027, 0.033, 0.030; $\beta = 0.5$: 0.011, 0.010, 0.010 (for $2I$, Q_8, $SU(2)$). The file records no error for m_{eff}, so R3c is scored without one.

Reading. This is not Lorentzian 3+1 physics: space and time are treated differently, and Q_8 has no continuum limit (l.240).

4.4 Gap 4: chirality (obstruction recorded)

No preregistration was frozen because no computation was run (l.245).

[P] `TOE-G4-O1` The core has no chirality: H_U is second order (no Dirac operator; mirror theorem); the hull is 3-dimensional (no γ_5 as Euclidean spacetime); on any finite patch a closed loop of $U(1)$ twists has zero net spectral flow (finite-volume content of Nielsen–Ninomiya); a domain wall or overlap would insert chirality, not derive it. unverified

Evidence: TOE_ATTEMPT.md §3.4 l.245-255; ledger.json TOE-G4-O1

[13]

[P] `TOE-G4-O2` An $SU(2)$ theory with a single Weyl doublet is inconsistent (Witten 1982); any chiral version of the $SU(2)$ θ-family needs an even number of doublets. unverified

Evidence: TOE_ATTEMPT.md §3.4 l.256; ledger.json TOE-G4-O2

[X] `G4-INFVOL` The infinite-volume statement for the hull (no single Weyl fermion) is not covered by the 1981 Nielsen–Ninomiya proofs, which assume translation invariance, and is not proved.

Evidence: TOE_ATTEMPT.md §3.4 l.251,258

Nielsen–Ninomiya are cited as pointers for the 1981 no-go theorem [19, 20] (abstracts only). The draft's "finite-volume content" reading cannot be checked from those abstracts; the infinite-volume, translation-invariant proofs do not cover the finite-patch claim as stated (G4-INFVOL). A domain wall inserts chirality rather than deriving it [9]; no verified citation for an overlap operator is inserted here. An $SU(2)$ theory with an odd number of left-handed doublets (and no other representations) is inconsistent [30] (abstract pointer; statement pages were not seen).

4.5 Gap 5: generations

The only threefold structure is the set of three $\mathbb{Z}_2$ shadows χ_1, χ_2, χ_3, the non-trivial characters of $V_4 = Q_8/\{\pm 1\}$ (l.264).

[P] `G5-S1` S1: an exact axis symmetry of the rule, $\text{conj}_Q(d) = d$, makes the shadow floors constant on the orbits of the symmetry group on the axes. unverified

Evidence: TOE_ATTEMPT.md §3.5 l.267

[P] `TOE-G5-S3` S3: the shadows are three different Q_8 characters; a symmetry permuting them acts as an outer automorphism of the gauge group, whereas generations carry identical gauge charges. unverified

Evidence: TOE_ATTEMPT.md §3.5 l.268; ledger.json TOE-G5-S3

[P] `G5-S4` S4: the only non-Abelian sector, the spinor, has multiplicity 2, and the Chair44 index is $2\chi = 2$. unverified

Evidence: TOE_ATTEMPT.md §3.5 l.269

[C] `G5-COUNT` Structural count among the 6561 proper rules: 6456 have no axis symmetry, 24 a symmetry group of order 2, 80 of order 3, and exactly one (rule 63) the full S_3.

[13] **Flag (TOE-G4-O1):** ledger evidence cites stale 'TOE_ATTEMPT.md s4.4' (now §3.4)

Evidence: TOE_ATTEMPT.md §3.5 l.271-275; src/rule_symmetry.py

[E] TOE-G5.1 G5.1 holds: rule 63 (S_3), $\lambda_1 = \lambda_2 = \lambda_3$ on $T_2 \dots T_5$ (7.868057, 5.345482, 4.678361, 4.280895), equal to 10^{-13} in floating point.

Evidence: TOE_ATTEMPT.md §3.5 l.281; ledger.json TOE-G5.1; source label [C]

[E] TOE-G5.2 G5.2 holds: rule 64 (C_3), the same (10.838889, 8.860622, 7.680477, 6.323312), equal to 10^{-13} in floating point.

Evidence: TOE_ATTEMPT.md §3.5 l.282; ledger.json TOE-G5.2; source label [C]

[E] TOE-G5.3 G5.3 holds: FR30 $\lambda_x = \lambda_z \neq \lambda_y$ (e.g. T_5: 7.700134, 5.386980, 7.700134), floating point.

Evidence: TOE_ATTEMPT.md §3.5 l.283; ledger.json TOE-G5.3; source label [C]

[E] TOE-G5.4 G5.4 holds: 1387, x539311, x539291 have three distinct floors (minimum pair gap 0.01486, 1387, T_4), floating point.

Evidence: TOE_ATTEMPT.md §3.5 l.284; ledger.json TOE-G5.4; source label [C]

[C] TOE-G5.5 G5.5 holds: conjugation by W_Q permutes $\chi_k \to \chi_{Q(k)}$ for all six Q.

Evidence: TOE_ATTEMPT.md §3.5 l.285; ledger.json TOE-G5.5

[P] TOE-G5-none No mechanism in the core produces three generations. unverified

Evidence: TOE_ATTEMPT.md §3.5 l.288; ledger.json TOE-G5-none

14

[A] G5-R63 The degenerate shadow triplet of rule 63 is related by an S_3 that permutes gauge charges; it is at most an analogue of a flavour or colour triplet, not three generations.

Evidence: TOE_ATTEMPT.md §3.5 l.289

4.6 Gap 6: gravity (obstruction recorded)

The two claims below are separate from the T1 gravity run. Where this note mentions that run: the $L = 1$ interior count is 0, not 6; $N_{\text{int}} = 6$ holds at $L = 2, \dots, 5$ [C]; no local tile-wise moduli, and $Q = 0$, are [C] only at $L = 1, \dots, 5$ in this tile-wise constant, tangentially continuous metric model; the all-level claim is [X]. This is not a classification of all Regge formulations. TOE-G6-O1 is an elementary geometry statement that Regge deficits vanish on the chair complex (a hand proof, unverified); it is not T1's "no local moduli" result, and "rigid" in the claim text should not be read that way. "Regge calculus" is used here as a name [24] (pointer only; the deficit-angle definition and the action were not seen from the abstract), not as support for the chair-complex deficit-zero calculation. TOE-G6-O2 (that dynamical geometry leaves the substitution class) is a one-line remark in TOE_ATTEMPT.md §3.6, also unverified. Causal dynamical triangulations are a standard way to make the geometry dynamical [1]; that reference does not support the claim that doing so leaves the substitution class or destroys legal-patch structure.

[P] TOE-G6-O1 The substitution geometry is flat and rigid: every interior lattice edge lies in exactly four unit cubes, so every Regge deficit angle is $2\pi - 4 \cdot \frac{\pi}{2} = 0$ and the Regge action vanishes at every level; the substitution is an exact similarity by a factor 2. unverified

Evidence: TOE_ATTEMPT.md §3.6 l.297-300; ledger.json TOE-G6-O1

15

[P] TOE-G6-O2 Making the geometry dynamical (Regge calculus with variable edge lengths, or causal dynamical triangulations) leaves the substitution class and destroys the legal-patch structure used by K2–K10. unverified

[14] **Flag (TOE-G5-none):** ledger evidence cites stale 'TOE_ATTEMPT.md s4.5' (now §3.5)
[15] **Flag (TOE-G6-O1):** ledger evidence cites stale 'TOE_ATTEMPT.md s4.6' (now §3.6)

Evidence: TOE_ATTEMPT.md §3.6 l.301; ledger.json TOE-G6-O2
16

5 Scorecard after Phase 3

Table 3: Scorecard (TOE_ATTEMPT.md §4 l.310–321, identical to SCORECARD.md). Partial-result labels in rows 3 and 4 are restated with the paper labels of Appendix C; the source prints "[P]/[C]" for row 3 and "one continuum trajectory [E]" for row 4. The [P] entries are hand proofs [P, unverified].

#	requirement	status	partial results
1	$SU(3) \times SU(2) \times U(1)$	missing	$SU(2)$ as dynamical field [E]
2	chiral fermions	missing	obstruction (G4)
3	three generations	missing	obstruction (G5): [P] S3; [C] symmetry count; [E] floor equalities
4	Lorentzian 3+1 and continuum limit	missing	lattice time direction [P]; one θ-trajectory, $L \leq 6$ [E]; its continuum limit [X]
5	general relativity in some limit	missing	obstruction [P] (G6)
6	QM with unitary time evolution	[P] lattice level only	G3 R0 (standard)
7	dimensionless constants, derived	missing	model constants derived [E]
8	(internal) floor as order parameter	[P] negative	G1 T3

Rows open: 1, 2, 3, 4, 5, 7.[17]

6 What would falsify this

Each label comes with its own refutation route (TOE_ATTEMPT.md §5 l.324–339).

- **[P] claims** (hand proofs: the diamagnetic bound and compression T1, T2, the infinite-volume statement T3, the transfer-matrix positivity R0, the obstructions of G4–G6, Theorems A–F) are refuted by a counterexample or by an error in the argument. A finite-group Wilson ensemble with a positive infinite-volume floor at some β would refute T3; a primitive frame rule for which c_L does not converge would refute Theorem F.
- **[C] claims** are refuted by an independent recomputation that disagrees, or by an error in the rounding analysis on which the certificates rest. A value of c_L for FR30 or 1387 at some large L outside the certified bracket (beyond the finite-L factor of Theorem C) would refute the bracket or one of Theorems B, C, F.
- **Preregistered [E] predictions** are refuted by reruns outside the frozen bands: new seeds, larger levels, more chains, or an independent implementation. The frozen protocols and seeds are listed in Appendix D.

[16] **Flag (TOE-G6-O2):** TOE_ATTEMPT item 2 carries no explicit inline label; label [P] taken from ledger TOE-G6-O2 (section-level 'each statement' rule only in §3.4)

[17] The source README.md says "One scorecard row is [P]"; SCORECARD.md has row 6 [P] and row 8 [P] negative (discrepancies.md §2.7). Row 8 is programme-internal, not a physics requirement.

- **[X] claims**: the continuum-limit conjectures G2B-LIMIT and G2B-MASS would be refuted by levels $L \geq 7$ whose c_L leave the certified bracket or stop converging geometrically, or by a non-vanishing higher-order term that survives $\theta = \kappa a \to 0$.

The programme as a route to interacting physics is already close to refuted. If the θ-trajectory limit is free for every rule and every κ, the core carries no interacting continuum physics. That general statement is open; it has been checked only for FR30 and 1387 and only at second order in θ.

7 Novelty and relation to prior work

No result here is claimed new except the elementary G5 statement about Chair44 rules (G5-COUNT [C]) and the specific application of homogenisation described in §7.1 (TOE_ATTEMPT.md §6 l.341–355).

- G1 T3: the finite-energy and Borel–Cantelli write-up is the author's and is not claimed new. Nakamura records $\inf \sigma = 0$ almost surely in a different model [18] and is not cited as the source of that argument.
- G1 H1/H2: generic pointer for Poisson statistics [17] (Minami treats bulk local statistics of the i.i.d. Anderson model, not ground-state or extreme-value laws for random Q_8 connections on nested supertiles; those claims are already [F]). No novelty is claimed.
- $2I$ freezing: first-order / freezing behaviour for $2I$ is recorded in [23] (abstract only). That paper is not cited for Q_8 or for the spherical-design argument, which were not seen in the abstract.
- G2 Part A self-energy: standard mechanism, no new method. No verified weak-coupling reference for the C/β self-energy is in the bibliography, and none is inserted here.
- G3 R0, G4, G6: standard (citations as in §4.3, §4.4, §4.6).

7.1 Related work: homogenisation and Theorem F

Theorem F is an application of known ergodic (stochastic) homogenisation results to the framed-chair hull; it is not a new limit theorem. A novelty check of Theorem F (research-publishing/novelty/theorem-F-novelty.md, 2026-10-31, status PARTIAL: the search stopped early under a rate limit, and Semantic Scholar and Google Scholar were not queried) reaches the following conclusions, which this section adopts.

- **The corrector formula is standard.** $\rho(d) = 7 \min_{\xi \in \mathcal{C}} E_\mu \sum_k |w_k + \xi_k|^2$, with minimiser $-P_{\mathcal{C}} w$, is the discrete corrector (homogenised-coefficient) formula of Kozlov [12] and of Künnemann [13] (not written there in min / $-P_{\mathcal{C}} w$ form). The continuum variational form is Kozlov's 1979/80 averaging paper [11]; the discrete formula is the 1986/87 difference-scheme paper. A continuum formula of Papanicolaou and Varadhan was cited in an earlier draft; that paper is not in the verified bibliography and is not cited here.
- **Convergence of periodised approximations is known in ergodic media.** Convergence of periodised effective coefficients to the homogenised one is proved by Owhadi [22]. Convergence of more general finite-box approximations was previously attributed to Bourgeat and Piatnitski; that paper is not in the verified bibliography and is not cited here.
- **Homogenisation on aperiodic tilings is known.** Braides, Riey and Solci homogenise energies on Penrose tilings [6]. Telcs proves an invariance principle on the Penrose tiling by an environment-process method [27]; the corrector method on that tiling is Bartha and Telcs [2].
- **Deterministic and survey references.** Biskup's survey of the random conductance model [4] is the standard reference for the discrete corrector, and his work on deterministic conductances [5] treats homogenisation without randomness.

- **The θ^2 link.** That the second-order coefficient of the lowest eigenvalue of the twisted operator equals a homogenised coefficient (Theorems A and F together) is analogous to Bloch-wave homogenisation [A] [7] (metadata verified; content not checked).

What is specific to this note, and possibly new only in combination (novelty note §3, verdict "partially anticipated"): (i) the finite-volume problems are Perron-weighted problems on pinned supertiles T_L with free boundary, rather than periodised or Dirichlet boxes; (ii) the limit is deterministic along the single sequence T_L, the ergodic input being unique ergodicity of the substitution hull [25, 26] rather than an average over a random environment (unique ergodicity of this hull remains an unverified input; see the scope remarks in §4.2); (iii) the θ^2 link above. No rate of convergence is proved; a quantitative rate for $|c_L - \rho|$ would go beyond the cited literature and would tighten the certified brackets. Theorem F also uses a multiparameter ergodic theorem [28] and Kolmogorov–Riesz compactness [8]; the Wiener citation is a documented cube-versus-ball gap, not a newly verified input (see §4.2). Dorodnyi (arXiv:2005.06516, 2020) treats periodic homogenisation, including periodic magnetic examples; it does not cover the substitution hull, so it does not apply to G2 Part B or Theorem F.

8 Where it stands

The certified mathematics of the core stands: the results K1–K15 and the certified numbers of Corollaries D and E do not depend on any physics reading. What Phase 3 adds is a set of lattice-level facts and one conditional limit statement.

- For finite gauge groups the matter floor is not an infinite-volume order parameter (T3, SC-8).
- In the dynamical $SU(2)$ theory the gauge-induced floor shift is the perturbative self-energy, to 1 % at $\beta = 64$ (G2A-SELFENERGY [E]); $2I$ and Q_8 freeze as designed, Q_8 earlier than predicted (W3c [F]).
- Adding a time direction gives a positive transfer matrix and the derived strong-coupling spectrum (R0, G3-SPECTRUM), but not Lorentzian 3+1 physics.
- Along the θ-trajectory the second-order coefficient converges (Theorem F, for primitive rules) and is certified to lie in an interval of width about 2.23 for FR30 and 1387 (Corollary E, conditional on Theorems B, C, F); numerically it behaves like a mass term for $L \leq 6$ [E], and the continuum limit is a conjecture [X].

Every limit found so far looks like free physics, and the substitution hull acts as an inert lattice (TOE_ATTEMPT.md 1.390). Chirality, generations and gravity carry recorded obstructions, not partial successes.

9 Phase summaries

Phase 0 labelled 1243 claims from the work. Of these, 327 are excluded (the excluded lineage, together with every claim already marked [F]); 139 rest on hard-coded tests; and 192 Lean theorems whose names promised more than they proved were relabelled. All eight Chair44 certificate files were rechecked by the Lean 4.35.0 kernel and use only standard axioms. Phase 0 itself recorded no new failure; the audit confirms earlier refutations already marked [F].

Phase 1 stated the core object in CORE.md and listed the proved and certified properties K1–K15. What fits is the Chair44 work, the Tseitin $\mathbb{Z}_2$ coboundary results, and the Unique Games reading; what does not fit is random 3-SAT, solver and ML engineering, and the 6D→3D icosahedral modules.

The Unique Games reading (K11) is the source's recommended bridge between the complexity work and the floor.

Phase 2 recorded a baseline in which every physics scorecard row was missing, with seven no-go facts each traced to a source. The recommended first attacks were the volume question and the continuum question.

Phase 3 ran the preregistered gap attacks. Row 6 became [P] at the lattice level (G3). Row 4 gained a lattice time direction [P] and one θ-trajectory at finite L [E]; identifying that trajectory's limit as a continuum or mass term remains [X] in this note (owner ledger sync, 6 November 2026). Row 8 was settled negatively [P]. Rows 2, 3 and 5 now carry explicit obstructions: for row 3 the obstruction itself is [P] (S1/S4), the symmetry count is [C], and the numerical degeneracies (G5.1–G5.4) are [E]. The Phase 3 failures, all recorded as [F], were G1 H1 (weakest-child localisation), G1 H2 (min-stable law at $L = 3, 4$), G1 H3b (Q_8 significantly below $SU(2)$), G2 W3c (Q_8 tracks $SU(2)$ at $\beta \leq 1$), and G2 B4 (a uniform shift of 20 levels at $L = 5$; a post-hoc diagnostic points to an eigensolver artifact). Other preregistered items were not falsified in Phase 3. The recommended next step was to prove the G2 Part B homogenisation limit.

After Phase 4 that step was written in PROOF.md. No scorecard row moved, because the limit is free physics. What moved from [E] to a written proof [P, unverified] is the existence of $\lim c_L$ for primitive rules (1 666 486 rules), its identification as a hull corrector formula, and the uniform lower bound 3527/1012 for all 6^8 rules, together with certified brackets [C] for FR30 and 1387. Identifying those finite-level constants as a continuum mass term remains [X]. Still open are the θ^3 remainder along the trajectory (B3 stays [E]); the 13 130 imprimitive rules, including x539311 and x539291, whose hull has two components and whose even- and odd-level convergence is open; and machine checking or independent review of the written proof. The source notes that a sloped or periodic cell problem would be a cheap way to tighten the upper end of the bracket, whose width is about 2.23 at $\ell = 6$.

10 Figures

Figures 1–4 are identical to the frozen v1.0 set (no $\ell = 7$ series), built from printed PROOF.md values or stored result files. Figure 5 was rebuilt for this draft from the outward enclosure [7.954291, 8.7642270382] [E]. The set lives in `figs_v0.2/`; regenerate Figure 5 with `make_figures.py -paper` (venv: matplotlib 3.11.2, numpy 2.5.4) after copying Figures 1–4 from `figs_v1.0/`. Figures 1–4 plot the values *as printed in PROOF.md*, including the ten nearest-rounded cells of Table 1, which they flag. Labels inside the figure legends follow the source; where they differ from the paper labels, the caption gives the paper label, which applies.

[E] `FIG5-ESTAR` $E_*(\mathrm{FR30}) \in [7.954291, 8.7642270382]$, rounded outward from the two lower-end certificates and the exact rational upper bound (ledger only, not in PROOF.md).

Evidence: ledger.json CH9-N3-enclosure (l.624), CH8-B1-upper-bound (l.474); figures/proof-brackets/README.md Fig.5; source label [C]

[19] Upper end: the exact certified bound is 95969095435322528/10950092349041091 = 8.764227038115…, so the outward 10-decimal value is 8.7642270382 (discrepancies.md §1). Lower end: there are two independent certificates, 7.954292265 and 7.954291278 (discrepancies.md §2.3). The ledger's 7.954292 is below the first but above the second, so it is not safe with respect to both; this note prints 7.954291, which is below both. The paper label is [E] until ratification of the Phase IX mapping that sent the source category to [C] (discrepancies.md §2.2). The owner first pass of 6 November 2026 re-checked this outward arithmetic [C] and did not re-derive the certificates from source files. The source files are unchanged.

[20] **Flag (FIG5-ESTAR):** v0.0/v0.1 CSV short text printed the ledger interval [7.954292, 8.7642270381] (truncated upper). v0.2 paper label [E] until Phase IX mapping is ratified, and short text [7.954291, 8.7642270382], matching the outward paper statement. Owner first pass 2026-11-06 re-checked the outward arithmetic [C] and did not re-derive the

11 Limitations and open problems

- **Not a theory of everything.** Six of seven physics scorecard rows are open; the only [P] physics row (QM with unitary evolution) holds at lattice level and is a standard construction.
- **Every limit found looks like free physics.** Along the θ-trajectory, for $L \leq 6$, the second-order coefficient behaves like a mass term [E]; that a continuum limit exists and is a free massive field is a conjecture [X]. The $SU(2)$ regime of Part A is ordinary weak-coupling lattice gauge theory [E].
- **Unverified proofs.** The PROOF.md argument (Theorems A–F, Lemmas 6 and 7, the rounding analysis) and the G1/G3/G4/G5/G6 [P] statements are hand proofs, not machine-checked and not peer-reviewed. The certified brackets and the bound 3527/1012 depend on them. An external expert reading or a formalisation is needed before any of these loses its unverified tag.
- **Open mathematics.** Uniformity of the θ^3 remainder along $\theta_L = \pi 2^{-L}$ (B3 remains [E]); the 13 130 imprimitive rules (even/odd-level limits); the infinite-volume chirality statement for the hull [X]; whether the θ-trajectory limit is free for *every* rule and κ.
- **Falsified predictions.** G1 H1, H2, H3b; G2 W3c, B4; Phase XIV H1, H2; Phase VI gapped claim; QIS-299. B4's post-hoc diagnostic does not change its [F]; a new frozen protocol is needed.
- **Finite sizes and statistics.** G1 up to T_5 only (the rare regions are estimated to appear near 10^{22} tiles, G1-RARE [X]); G2 A has two chains per point and 200 measurements, so autocorrelations are handled only through the 5-sweep spacing; G2 B has six levels, and its c_∞ depends on the fitting range by up to 0.10. Standard errors are given where the sources record them (G1 floors and H3, Table 2, G3 R3a); the G3 R2 standard errors are taken from results/g3_score.json (Section 4.3); the G3 R3c effective masses have no recorded error; seeds and run counts are in Appendix D.
- **Bounds hygiene.** Ten PROOF.md table cells are rounded to nearest, the ledger's E_* upper end 8.7642270381 is truncated, and its lower end 7.954292 is above one of the two certificates; this note prints outward values (Table 1, Appendix C). PROOF.md prints 15.465590 and the ledger 15.465589; both are safe. The source files are unchanged.
- **Label mismatches with the ledger** (Appendices B and C): resolved here toward the weaker label. As of the 6 November 2026 owner ledger sync, the previously unmapped rows now use the paper id as the ledger id (recorded in claims.csv). Two paper labels, G1-RARE [X] and G2B-DERIV [E], still carry a pending author ruling on the source's non-standard P-level wording; no ruling is invented here. K6 keeps printing [C] as the weakest component of a compound whose X4 no-go remains [P] in the ledger.
- **Reproducibility.** The source was read from a file snapshot; Appendix D gives file hashes, seeds and parameters, but no commit hashes. [TODO: toe repository commit and prereg commits: needed from the author]

A Status of claims

Total labelled claims: 99. By paper label: [P] 33, [C] 12, [E] 33, [X] 5, [A] 5, [F] 11. By source label (verbatim, including non-standard and compound labels): [A] 5, [C] 15, [E] 26, [E, post hoc] 1, [F] 11,

certificates from source files

[P] 34, [P+C] 2, [P-level] 1, [P-level estimate] 1, [P/C] 1, [X] 2. 15 claims carry a paper label different from their source label (Appendix C); 2 carry a pending-ruling marker. Of the claims printed [P], 32 are hand proofs marked unverified.

ID	Where	Label	Statement (short)	Evidence
S1-QIS-PNP	§1	[F]	QIS 'unconditional P != NP' claim is refuted by the repo's own audit	TOE_ATTEMPT.md §8 l.401; CORE.md §5 l.84; ledger.json QIS-322 [F, excluded as falsified]
S1-QIS-PLANNER	§1	[F]	QIS '64-0 planner' record is refuted by the repo's own audit	TOE_ATTEMPT.md §8 l.401; CORE.md §5 l.84; ledger.json QIS-182 [F, excluded as falsified]
K1	§2.2	[P] unverified	flat <=> gauge-trivial <=> lambda_min(H_U)=0 on finite connected graphs	CORE.md §2 l.38 (source: Chair44 X1)
K2	§2.2	[P] unverified	E_* = inf over legal patches of Dirichlet floor; depends only on legal environments	CORE.md §2 l.39 (source: XII.A)
K3	§2.2	[P] unverified	two-sided IMS localisation with optimal R^-2 rate	CORE.md §2 l.40 (source: XII.B, XII.C)
K4	§2.2	[P]	forced curvature: >=12 of 21 centre-octant two-cycles non-trivial (Lean checks encoded finite predicates only; geometric statement not kernel-checked)	CORE.md §2 l.41 (source: Phase VII-B P1)
K5	§2.2	[C]	all 1098 gauge classes of the 6561 proper rules have E_* >= 3.788	CORE.md §2 l.42 (source: X3)
K6	§2.2	[C] unverified source: P/C (App. C)	plaquette-class frequencies give no lower bound on E_*	CORE.md §2 l.43 (source: X4, X5, XII.P63); source label [P/C]
K7	§2.2	[P] unverified	non-spinor floors are functions of the plaquette class field	CORE.md §2 l.44 (source: XI.H3)
K8	§2.2	[P] unverified	Orbit Lemma	CORE.md §2 l.45 (source: XII.OL)
K9	§2.2	[C]	certified E_* enclosures for 1387 and 539311; maximiser among three rules	CORE.md §2 l.46 (source: XII.R20, XII.r24, XII.F4)
K10	§2.2	[C] unverified source: P+C (App. C)	no function of level-one data determines E_*	CORE.md §2 l.47 (source: XII.T31); source label [P+C]
K11	§2.2	[P] unverified	floor bounded by unique-games frustration energy of any gauge section	CORE.md §2 l.48 (source: XII.ug.P151)
K12	§2.2	[P] unverified	Clifford transports lie in Q8; adjoint image V4; Z4-valued in 2D	CORE.md §2 l.49 (source: CLIFFORD_Q8_PHASE2 Thms 2.2-2.3; X6)
K13	§2.2	[P] unverified	Wilson traces and plaquette classes are mirror-invariant	CORE.md §2 l.50 (source: CLIFFORD_Q8_PHASE2 Thm 2.4)
K14	§2.2	[P] unverified	diamagnetic bound (Kato's inequality)	CORE.md §2 l.51 (source: standard; prereg/G1_volume_scaling.json T1)

ID	Where	Label	Statement (short)	Evidence
K15	§2.2	[P] unverified	infinite-volume floor 0 a.s. for beta=0 and finite-beta finite-group Gibbs measures	CORE.md §2 l.52 (source: Phase VII §8; prereg/G1_volume_scaling.json T3)
I-TSEITIN	§2.3	[P] unverified	Tseitin formulas unsat iff total charge odd; Z2 coboundary side of the core	TOE_ATTEMPT.md §2 l.63; CORE.md §3 l.61
I-ENERGY	§2.3	[P] unverified	multilinear energy of a Tseitin instance has floor >= 1	CORE.md §3 l.62
I-ENERGY-INST	§2.3	[A]	Tseitin energy floor as an instance of the core is analogy only	CORE.md §3 l.62
I-XIII-SCOPE	§2.3	[P] unverified	scope: external UG theorem is worst-case on finite instances and bounds no floor	TOE_ATTEMPT.md §2 l.64; CORE.md §3 l.60
I-UG-EXT	§2.3	[X]	external UG / 'ETH is false' claims recorded as external, unchecked	CORE.md §3 l.69
I-QIS320*	§2.3	[E]	lambda_2 of SAT clause graphs plateaus near 1.01	CORE.md §3 l.64; ledger.json QIS-320
I-QIS320-INST	§2.3	[A]	QIS lambda_2 as an instance of the core is analogy only	CORE.md §3 l.64; §5 l.87
I-QIS299	§2.3	[F]	QIS-299 'frustrated clauses carry a non-trivial 1-cycle' is falsified	CORE.md §3 l.65; ledger.json QIS-299
I-GU-C4	§2.3	[C]	4D chair C4 face counts, chi=1; boundary chi=0	CORE.md §3 l.67; ledger.json GU-001
I-GU-A	§2.3	[A]	Geometric Unity: structural analogy only	CORE.md §3 l.68
I-WOLFRAM	§2.3	[A]	Wolfram hypergraph rewriting: analogy only	CORE.md §3 l.70
I-XIV-H1	§2.3	[F]	Phase XIV H1 (hot-series floor non-increasing) falsified	CORE.md §3 l.59; TOE_ATTEMPT.md §4 row 8 l.319; ledger.json CH14-XIV.H1
I-XIV-H2	§2.3	[F]	Phase XIV H2 falsified	CORE.md §3 l.59; TOE_ATTEMPT.md §4 row 8 l.319; ledger.json CH14-XIV.H2
I-XIV-H3	§2.3	[E]	Phase XIV H3 holds	CORE.md §3 l.59; ledger.json CH14-XIV.H3
I-VI-017	§2.3	[F]	Phase VI 'random Q8 links are gapped (14.4)' withdrawn/falsified	TOE_ATTEMPT.md §3.1 l.115; ledger.json CH-VI-017
G1-T1	§3.1	[P] unverified	T1: lambda_1(T_L) <= lambda(T_L) <= min over 8 children	TOE_ATTEMPT.md §3.1 l.90
G1-T2	§3.1	[P] unverified	T2: P(lambda(T_L)>x) <= P(lambda(T_{L-1})>x)^8	TOE_ATTEMPT.md §3.1 l.91
G1-T3*	§3.1	[P] unverified	T3: inf spec H_U = 0 a.s. at every beta (finite groups)	TOE_ATTEMPT.md §3.1 l.92-94; ledger.json TOE-G1-T3
G1-RARE*	§3.1	[X] source: P-level estimate (App. C) pending ruling	rare regions first appear at ~1e22 tiles	TOE_ATTEMPT.md §3.1 l.95; source label [P-level estimate]
G1-FLOORS	§3.1	[E]	Q8 mean floors T1..T5: 18.511 ... 14.367	TOE_ATTEMPT.md §3.1 l.98; results/g1_score.json
G1-TRIVIAL	§3.1	[E]	trivial floor 11.21 -> 0.139	TOE_ATTEMPT.md §3.1 l.99
TOE-G1-T1*	§3.1	[E]	every beta=0 sample obeys T1	TOE_ATTEMPT.md §3.1 l.100; ledger.json TOE-G1-T1 [C]

ID	Where	Label	Statement (short)	Evidence
TOE-G1-T2	§3.1	[E]	T2 holds at every level (data check)	TOE_ATTEMPT.md §3.1 l.100; ledger.json TOE-G1-T2
TOE-G1-H1	§3.1	[F]	H1 weakest-child localisation	TOE_ATTEMPT.md §3.1 l.106; ledger.json TOE-G1-H1
TOE-G1-H2	§3.1	[F]	H2 min-stable median law	TOE_ATTEMPT.md §3.1 l.107; ledger.json TOE-G1-H2
TOE-G1-H3a	§3.1	[E]	H3a 2I and SU(2) agree	TOE_ATTEMPT.md §3.1 l.108; ledger.json TOE-G1-H3a
TOE-G1-H3b	§3.1	[F]	H3b Q8 significantly below SU(2)	TOE_ATTEMPT.md §3.1 l.109; ledger.json TOE-G1-H3b
TOE-G1-H4	§3.1	[E]	H4 medians above certified E_*(1387) (weak test)	TOE_ATTEMPT.md §3.1 l.110; ledger.json TOE-G1-H4
SC-8*	§3.1	[P] unverified	row 8: floor is not an infinite-volume order parameter ([P] negative)	TOE_ATTEMPT.md §3.1 l.116; §4 row 8 l.319
G2B-DERIV*	§3.2 B	[E] source: P-level (App. C) pending ruling	second-order expansion of lambda(T_L,theta)	TOE_ATTEMPT.md §3.2 l.129; source label [P-level]
G2B-CL	§3.2 B	[E]	c_L table and geometric c_inf (FR30 13.364, 1387 14.285)	TOE_ATTEMPT.md §3.2 l.124,132-135; results/g2_partB.json
TOE-G2-B0	§3.2 B	[E] source: C (App. C)	B0 Taylor check	TOE_ATTEMPT.md §3.2 l.124,139; ledger.json TOE-G2-B0; source label [C]
TOE-G2-B1	§3.2 B	[E]	B1 geometric convergence	TOE_ATTEMPT.md §3.2 l.140; ledger.json TOE-G2-B1
TOE-G2-B2	§3.2 B	[E]	B2 c_6>0 and c_inf>0	TOE_ATTEMPT.md §3.2 l.141; ledger.json TOE-G2-B2
TOE-G2-B3	§3.2 B	[E]	B3 scaling trajectory (remains [E])	TOE_ATTEMPT.md §3.2 l.142,153; PROOF.md §0 not-proved item 1; ledger.json TOE-G2-B3
TOE-G2-B4	§3.2 B	[F]	B4 uniform shift of 20 levels	TOE_ATTEMPT.md §3.2 l.143; ledger.json TOE-G2-B4
G2B-LIMIT	§3.2 B	[X] source: E (App. C)	continuum limit along theta = kappa a	TOE_ATTEMPT.md §3.2 l.146; §7 l.371; source label [E]
G2B-MASS	§3.2 B	[X] source: E (App. C)	frustration survives as mass term kappa^2 c_inf	TOE_ATTEMPT.md §3.2 l.147; §4 row 7 l.318; §7 l.372; source label [E]
G2B-POSTHOC	§3.2 B	[E] source: E, post hoc (App. C)	post-hoc B4 diagnostic: eigensolver artifact plausible, not established	TOE_ATTEMPT.md §3.2 l.159-164; src/posthoc_b4.py; source label [E, post hoc]
TOE-PROOF-A	§3.2 PROOF	[P] unverified	Theorem A: second-order expansion and variational formula for c_L	PROOF.md §0 l.10, §2; ledger.json TOE-PROOF-A
TOE-PROOF-B	§3.2 PROOF	[P] unverified	Theorem B: monotone densities mu_l, nu~_l	PROOF.md §0 l.11, §3; ledger.json TOE-PROOF-B
TOE-PROOF-C	§3.2 PROOF	[P] unverified	Theorem C: two-sided bracket of c_L	PROOF.md §0 l.12, §4; ledger.json TOE-PROOF-C
TOE-PROOF-D	§3.2 PROOF	[C] unverified source: P+C (App. C)	D: liminf c_L >= 3527/1012 for all 6^8 rules	PROOF.md §0 l.13, §5; ledger.json TOE-PROOF-D [C]; results/proof_positivity.json; source label [P+C]

ID	Where	Label	Statement (short)	Evidence
TOE-PROOF-E-FR30*	§3.2 PROOF	[C]	E FR30 bracket [13.232574, 15.465590]	PROOF.md §0 l.14, §5 l.184; ledger.json TOE-PROOF-E-FR30
TOE-PROOF-E-1387	§3.2 PROOF	[C]	E 1387 bracket [14.161980, 16.390966]	PROOF.md §0 l.14, §5 l.185; ledger.json TOE-PROOF-E-1387
PROOF-E-TABLE*	§3.2 PROOF	[C]	certified mu_l / nu0_l table l=1..6 (10 cells flagged)	PROOF.md §5 l.173-182; results/proof_bounds.json
TOE-PROOF-prim	§3.2 PROOF	[C]	primitivity holds for 1666486 rules, fails for 13130	PROOF.md §6 l.194-195; ledger.json TOE-PROOF-prim; src/proof_primitivity.py
TOE-PROOF-F	§3.2 PROOF	[P] unverified	Theorem F: c_L -> rho(d) corrector formula for primitive rules	PROOF.md §0 l.15, §6; ledger.json TOE-PROOF-F
PROOF-L7	§3.2 PROOF	[P] unverified	Lemma 7 (sublinearity of the integrated corrector)	PROOF.md §6 l.221-225
TOE-G2-D1*	§3.2 A	[E] source: P (App. C)	D1: weak-coupling limit C_pert (formal perturbation theory)	TOE_ATTEMPT.md §3.2 l.175-177,203; ledger.json TOE-G2-D1; src/pert.py; source label [P]
TOE-G2-W1	§3.2 A	[E]	W1 SU(2) approaches C_pert/beta	TOE_ATTEMPT.md §3.2 l.195; ledger.json TOE-G2-W1
TOE-G2-W2	§3.2 A	[E]	W2 monotone approach	TOE_ATTEMPT.md §3.2 l.196; ledger.json TOE-G2-W2
TOE-G2-W3a	§3.2 A	[E]	W3a 2I tracks SU(2) at beta<=4	TOE_ATTEMPT.md §3.2 l.197; ledger.json TOE-G2-W3a
TOE-G2-W3b	§3.2 A	[E]	W3b 2I frozen at beta=32,64	TOE_ATTEMPT.md §3.2 l.198; ledger.json TOE-G2-W3b
TOE-G2-W3c	§3.2 A	[F]	W3c Q8 tracks SU(2) at beta<=1	TOE_ATTEMPT.md §3.2 l.199; ledger.json TOE-G2-W3c
TOE-G2-W4	§3.2 A	[E]	W4 2I within 25% of SU(2) at beta=8	TOE_ATTEMPT.md §3.2 l.200; ledger.json TOE-G2-W4
G2A-SELFENERGY	§3.2 A	[E]	floor shift = perturbative self-energy to 1% at beta=64	TOE_ATTEMPT.md §3.2 l.203-205; §7 l.369
TOE-G3-R0	§3.3	[P] unverified	R0 positive self-adjoint transfer matrix (standard)	TOE_ATTEMPT.md §3.3 l.217-220; ledger.json TOE-G3-R0
TOE-G3-R1	§3.3	[E]	R1 positive log-convex correlators	TOE_ATTEMPT.md §3.3 l.231; ledger.json TOE-G3-R1
TOE-G3-R2	§3.3	[E]	R2 r(1)/u^2 within band	TOE_ATTEMPT.md §3.3 l.232; ledger.json TOE-G3-R2
TOE-G3-R3a	§3.3	[E]	R3a Q8 above SU(2) at beta=1	TOE_ATTEMPT.md §3.3 l.233; ledger.json TOE-G3-R3a
TOE-G3-R3b	§3.3	[E]	R3b 2I = SU(2)	TOE_ATTEMPT.md §3.3 l.234; ledger.json TOE-G3-R3b
TOE-G3-R3c	§3.3	[E]	R3c m_eff vs -2 ln u	TOE_ATTEMPT.md §3.3 l.235; ledger.json TOE-G3-R3c
G3-SPECTRUM	§3.3	[E]	strong-coupling spectrum matches derivation	TOE_ATTEMPT.md §3.3 l.238; §7 l.379
SC-6	§3.3	[P] unverified	row 6: QM with unitary evolution at lattice level	TOE_ATTEMPT.md §4 row 6 l.317; §4 l.321
TOE-G4-O1*	§3.4	[P] unverified	G4 obstruction: no chirality in the core	TOE_ATTEMPT.md §3.4 l.245-255; ledger.json TOE-G4-O1
TOE-G4-O2	§3.4	[P] unverified	Witten anomaly: even number of doublets needed	TOE_ATTEMPT.md §3.4 l.256; ledger.json TOE-G4-O2

ID	Where	Label	Statement (short)	Evidence
G4-INFVOL	§3.4	[X]	infinite-volume Nielsen-Ninomiya for the hull: open	TOE_ATTEMPT.md §3.4 l.251,258
G5-S1	§3.5	[P] unverified	S1 axis symmetry -> constant shadow floors on orbits	TOE_ATTEMPT.md §3.5 l.267
TOE-G5-S3	§3.5	[P] unverified	S3 shadows are different Q8 characters	TOE_ATTEMPT.md §3.5 l.268; ledger.json TOE-G5-S3
G5-S4	§3.5	[P] unverified	S4 spinor multiplicity 2, index 2chi=2	TOE_ATTEMPT.md §3.5 l.269
G5-COUNT	§3.5	[C]	6456/24/80/1 symmetry count (rule 63 unique S3)	TOE_ATTEMPT.md §3.5 l.271-275; src/rule_symmetry.py
TOE-G5.1	§3.5	[E] source: C (App. C)	G5.1 rule 63 threefold degenerate	TOE_ATTEMPT.md §3.5 l.281; ledger.json TOE-G5.1; source label [C]
TOE-G5.2	§3.5	[E] source: C (App. C)	G5.2 rule 64 threefold degenerate	TOE_ATTEMPT.md §3.5 l.282; ledger.json TOE-G5.2; source label [C]
TOE-G5.3	§3.5	[E] source: C (App. C)	G5.3 FR30 lambda_x = lambda_z != lambda_y	TOE_ATTEMPT.md §3.5 l.283; ledger.json TOE-G5.3; source label [C]
TOE-G5.4	§3.5	[E] source: C (App. C)	G5.4 three distinct floors for frontier rules	TOE_ATTEMPT.md §3.5 l.284; ledger.json TOE-G5.4; source label [C]
TOE-G5.5	§3.5	[C]	G5.5 conjugation permutes shadows	TOE_ATTEMPT.md §3.5 l.285; ledger.json TOE-G5.5
TOE-G5-none*	§3.5	[P] unverified	no mechanism for three generations	TOE_ATTEMPT.md §3.5 l.288; ledger.json TOE-G5-none
G5-R63	§3.5	[A]	rule-63 triplet is at most a flavour/colour analogue	TOE_ATTEMPT.md §3.5 l.289
TOE-G6-O1*	§3.6	[P] unverified	chair complex is flat and rigid (Regge deficits 0)	TOE_ATTEMPT.md §3.6 l.297-300; ledger.json TOE-G6-O1
TOE-G6-O2*	§3.6	[P] unverified	dynamical geometry leaves the substitution class	TOE_ATTEMPT.md §3.6 l.301; ledger.json TOE-G6-O2
FIG5-ESTAR*	§9	[E] source: C (App. C)	E_*(FR30) enclosure [7.954291, 8.7642270382] (outward)	ledger.json CH9-N3-enclosure (l.624), CH8-B1-upper-bound (l.474); figures/proof-brackets/README.md Fig.5; source label [C]

* flagged; see Appendix B.

B Label and ledger flags

- I-QIS320: ledger QIS-320 label E; source_label 'certified' (ledger did not upgrade)
- G1-T3: ledger evidence cites stale 'TOE_ATTEMPT.md s4.1' (now §3.1)
- G1-RARE: non-standard source label 'P-level estimate'; printed [X] pending ruling
- TOE-G1-T1: LABEL MISMATCH: TOE_ATTEMPT lists this under 'Results [E, simulated]', ledger says [C]; weaker [E] used (no upgrade); needs ledger reconciliation
- SC-8: README.md says 'One scorecard row is [P]' but row 8 is '[P] negative' (discrepancies.md §2.7)

- `G2B-DERIV`: non-standard source label 'P-level'; printed [E] pending ruling; the proved version is TOE-PROOF-A
- `TOE-PROOF-E-FR30`: ledger prints upper end 15.465589, PROOF.md 15.465590; stored value 15.465588903524477, so both are outward-safe. PROOF.md value kept
- `PROOF-E-TABLE`: v0.0 printed PROOF.md's values; 10 cells were nearest-rounded (discrepancies.md §1). v0.1 prints outward values from results/proof_bounds.json; PROOF.md values are given in table footnotes
- `TOE-G2-D1`: ledger [P]; TOE text: 'at the level of formal perturbation theory'; printed [E] (Appendix C)
- `TOE-G4-O1`: ledger evidence cites stale 'TOE_ATTEMPT.md s4.4' (now §3.4)
- `TOE-G5-none`: ledger evidence cites stale 'TOE_ATTEMPT.md s4.5' (now §3.5)
- `TOE-G6-O1`: ledger evidence cites stale 'TOE_ATTEMPT.md s4.6' (now §3.6)
- `TOE-G6-O2`: TOE_ATTEMPT item 2 carries no explicit inline label; label [P] taken from ledger TOE-G6-O2 (section-level 'each statement' rule only in §3.4)
- `FIG5-ESTAR`: v0.0/v0.1 CSV short text printed the ledger interval [7.954292, 8.7642270381] (truncated upper). v0.2 paper label [E] until Phase IX mapping is ratified, and short text [7.954291, 8.7642270382], matching the outward paper statement. Owner first pass 2026-11-06 re-checked the outward arithmetic [C] and did not re-derive the certificates from source files

C Label changes vs. source

Each row is a claim whose printed label differs from the label in its source. The rule is that a paper label may be weaker than the source label, never stronger. Labels marked "pending" await the author's ruling.

#	ID	Source label	Paper label	Reason
1	`K6`	[P/C]	[C]	compound source label [P]/[C]; printed as its weakest component
2	`K10`	[P+C]	[C]	compound source label [P]+[C]; weakest component; ledger CH12-XII.T31 is [C]
3	`G1-RARE`	[P-level estimate]	[X]	non-standard source label 'P-level estimate'; a heuristic order-of-magnitude estimate with no run near 1e22 tiles, so no numerical evidence: [X] *(label pending author ruling on 'P-level estimate')*
4	`G2B-DERIV`	[P-level]	[E]	non-standard source label 'P-level'; formal second-order expansion whose only support at this point is the floating-point B0 check; the proved statement is TOE-PROOF-A *(label pending author ruling on 'P-level')*
5	`TOE-G2-B0`	[C]	[E]	floating-point Taylor/finite-difference check without error control (referee §4)
6	`G2B-LIMIT`	[E]	[X]	existence of a continuum limit is not established by six levels; the frozen prereg itself labels the homogenisation step D3 '[X, derived]' (prereg/G2_scaling.json); uniformity of the theta^3 remainder is open (PROOF.md §0)
7	`G2B-MASS`	[E]	[X]	identifying the limit operator as Dirichlet Laplacian + kappa^2 c_inf rests on D3 [X] and on an extrapolated c_inf; the computed c_L values stay [E] (G2B-CL)
8	`G2B-POSTHOC`	[E, post hoc]	[E]	compound source label 'E, post hoc'; standard label [E], the qualifier 'post hoc, not preregistered' is kept in the statement

#	ID	Source label	Paper label	Reason
9	TOE-PROOF-D	[P+C]	[C]	compound source label [P]+[C]; weakest component; ledger TOE-PROOF-D is [C]
10	TOE-G2-D1	[P]	[E]	formal perturbation theory (TOE_ATTEMPT l.203); the asymptotic expansion is not proved; what supports the limit is the numerical agreement W1/W2 [E]
11	TOE-G5.1	[C]	[E]	floating-point eigenvalues, equality to 1e-13 without a rigorous error bound (TOE_ATTEMPT l.277; referee §4)
12	TOE-G5.2	[C]	[E]	floating-point eigenvalues, equality to 1e-13 without a rigorous error bound (TOE_ATTEMPT l.277; referee §4)
13	TOE-G5.3	[C]	[E]	floating-point eigenvalues, equality to 1e-13 without a rigorous error bound (TOE_ATTEMPT l.277; referee §4)
14	TOE-G5.4	[C]	[E]	floating-point eigenvalues; a minimum gap of 0.01486 without a rigorous error bound (TOE_ATTEMPT l.277; referee §4)
15	FIG5-ESTAR	[C]	[E]	source category mapped to [C] under a Phase IX mapping the author has not ratified (discrepancies.md §2.2; ledger CH9-N3); printed [E] until that mapping is ratified. Interval printed outward

Printed values changed vs. source. These are not label changes; they are listed for completeness. Table 1: ten cells rounded outward (printed PROOF.md values in the table footnote). E_*(FR30): upper end 8.7642270381 → 8.7642270382, lower end 7.954292 → 7.954291 (Figure 5 footnote). FR30 headline upper end: PROOF.md 15.465590 kept; ledger 15.465589 also safe.

D Reproducibility

Source snapshot. The toe repository was read from a file snapshot, not a git checkout; commit hashes are unknown. File hashes (sha256, first 8 hex digits): PROOF.md `dd992292`, TOE_ATTEMPT.md `2bb543b0`, CORE.md `cf1d5fd5`, SCORECARD.md `9aafae11`, ledger.json `46388050`, results/proof_bounds.json `b6752b18`. [TODO: toe commit hash and prereg commit hashes with timestamps]

Runs. Seeds are the `seed` fields of the frozen preregistrations (prereg/*.json), used by the scripts named.

Gap	script	seed	parameters and run counts
G1	src/g1_volume.py	20261029	$\beta = 0$, T_1–T_5; 2780 samples; Q_8: 400 per level at $L = 1$–3, 200 at $L = 4$, 60 at $L = 5$; $2I$, $SU(2)$: 200, 200, 200, 60
G2 A	src/g2_scaling.py	20261031	T_2, T_3; $\beta \in \{0.5, 1, 2, 4, 8, 16, 32, 64\}$; two chains (hot, cold); 500 thermalisation sweeps; 200 measurements every 5 sweeps
G2 B	src/g2_scaling.py, src/pert.py	n/a (deterministic)	FR30, 1387, $L = 1$–6; CG relative tolerance 10^{-12}
G3	src/g3_time.py	20261030	$T_2 \times \mathbb{Z}_8$; $\beta \in \{0.25, 0.5, 1\}$; 20 000 heat-bath measurements per run
G5	src/g5_generations.py	n/a (deterministic)	ARPACK `eigsh`, tolerance 10^{-13}
PROOF E	src/proof_bounds.py	n/a (deterministic)	FR30, 1387, $\ell = 1$–6; internal reruns (2026-10-31) of FR30 $\ell = 1$–6 and of both rules at $\ell = 1$–5 reproduced the stored bounds bit-identically; 1387 at $\ell = 6$ was not rerun

[TODO: software environment of the original runs (Python, NumPy, SciPy versions; platform): not recorded in the source]

References

[1] J. Ambjørn, J. Jurkiewicz, and R. Loll. Emergence of a 4d world from causal quantum gravity. *Physical Review Letters*, 93, 2004.

[2] Zs. Bartha and A. Telcs. Quenched invariance principle for the random walk on the penrose tiling, 2013.

[3] L. Bindi, P. J. Steinhardt, N. Yao, and P. J. Lu. Icosahedrite, Al63Cu24Fe13, the first natural quasicrystal. *American Mineralogist*, 96:928–931, 2011. Verified 2026-10-31 via Crossref.

[4] Marek Biskup. Recent progress on the random conductance model. *Probability Surveys*, 8:294–373, 2011.

[5] Marek Biskup. Homogenization theory of random walks among deterministic conductances. *Probability Theory and Related Fields*, 194:41–102, 2026. Issue year 2026 (PTRF 194); arXiv 2023; key kept as requested.

[6] Andrea Braides, Giuseppe Riey, and Margherita Solci. Homogenization of penrose tilings. *Comptes Rendus. Mathématique*, 347:697–700, 2009.

[7] Carlos Conca and Muthusamy Vanninathan. Homogenization of periodic structures via bloch decomposition. *SIAM Journal on Applied Mathematics*, 57:1639–1659, 1997. Metadata only; [A] link.

[8] Harald Hanche-Olsen and Helge Holden. The Kolmogorov-Riesz compactness theorem. *Expositiones Mathematicae*, 28:385–394, 2010. Verified 2026-10-31 via Crossref + arXiv.

[9] David B. Kaplan. A method for simulating chiral fermions on the lattice. *Physics Letters B*, 288:342–347, 1992.

[10] A. D. Kennedy and B. J. Pendleton. Improved heatbath method for monte carlo calculations in lattice gauge theories. *Physics Letters B*, 156:393–399, 1985.

[11] S. M. Kozlov. Averaging of random operators. *Mathematics of the USSR-Sbornik*, 37:167–180, 1980.

[12] S. M. Kozlov. Averaging of difference schemes. *Mathematics of the USSR-Sbornik*, 57:351–369, 1987.

[13] Rolf Künnemann. The diffusion limit for reversible jump processes on $\mathbb{Z}^d$ with ergodic random bond conductivities. *Communications in Mathematical Physics*, 90:27–68, 1983.

[14] Jeong-Yup Lee, Robert V. Moody, and Boris Solomyak. Consequences of Pure Point Diffraction Spectra for Multiset Substitution Systems. *Discrete and Computational Geometry*, 29(4):525–560, 2003. Verified 2026-11-07 via Crossref + arXiv API; Theorem 4.1 statement read in arXiv PDF.

[15] M. Lüscher. Construction of a selfadjoint, strictly positive transfer matrix for euclidean lattice gauge theories. *Communications in Mathematical Physics*, 54:283–292, 1977.

[16] Ryo Maezawa, Shiro Kashimoto, and Tsutomu Ishimasa. Icosahedral quasicrystals in Zn-T-Sc (T=Mn, Fe, Co, Ni) alloys. *Philosophical Magazine Letters*, 84:215–223, 2004. Verified 2026-10-31 via Crossref + arXiv.

[17] Nariyuki Minami. Local fluctuation of the spectrum of a multidimensional anderson tight binding model. *Communications in Mathematical Physics*, 177:709–725, 1996.

[18] S. Nakamura. Lifshitz tail for 2d discrete schrödinger operator with random magnetic field. *Annales Henri Poincaré*, 1:823–835, 2000.

[19] H. B. Nielsen and M. Ninomiya. Absence of neutrinos on a lattice. *Nuclear Physics B*, 185:20–40, 1981.

[20] H. B. Nielsen and M. Ninomiya. Absence of neutrinos on a lattice. *Nuclear Physics B*, 193:173–194, 1981.

[21] K. Osterwalder and E. Seiler. Gauge field theories on a lattice. *Annals of Physics*, 110:440–471, 1978.

[22] Houman Owhadi. Approximation of the effective conductivity of ergodic media by periodization. *Probability Theory and Related Fields*, 125:225–258, 2003.

[23] D. Petcher and D. H. Weingarten. Monte Carlo calculations and a model of the phase structure for gauge theories on discrete subgroups of SU(2). *Physical Review D*, 22:2465–2477, 1980.

[24] T. Regge. General relativity without coordinates. *Il Nuovo Cimento*, 19:558–571, 1961.

[25] Boris Solomyak. Dynamics of self-similar tilings. *Ergodic Theory and Dynamical Systems*, 17:695–738, 1997. Verified 2026-10-31 via Crossref.

[26] Boris Solomyak. Corrections to 'Dynamics of self-similar tilings'. *Ergodic Theory and Dynamical Systems*, 19:1685, 1999. Verified 2026-10-31 via Crossref.

[27] A. Telcs. Diffusive limits on the penrose tiling. *Journal of Statistical Physics*, 141:661–668, 2010.

[28] Norbert Wiener. The ergodic theorem. *Duke Mathematical Journal*, 5:1–18, 1939. Verified 2026-10-31 via Crossref.

[29] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10:2445–2459, 1974.

[30] Edward Witten. An su(2) anomaly. *Physics Letters B*, 117:324–328, 1982.

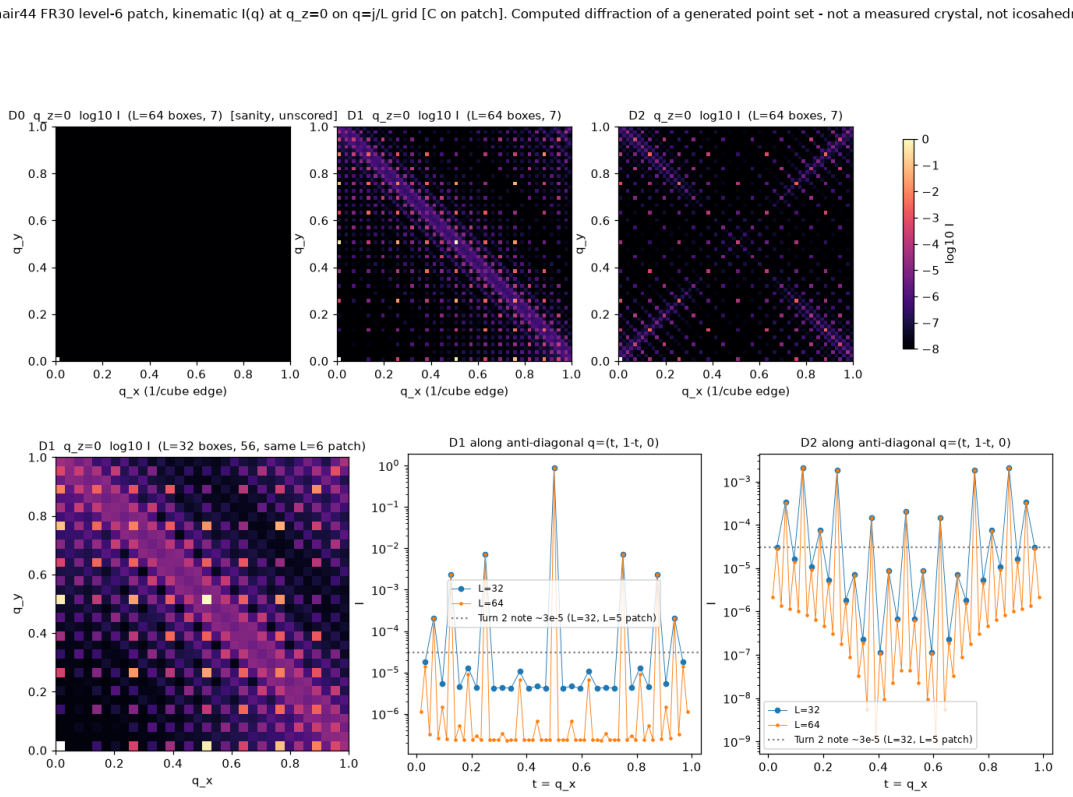


Figure 1: Chair44 FR30 level-6 patch: kinematic intensity on the $q_z = 0$ plane, $q = j/L$ (log scale), for decorations D0 (sanity), D1 and D2 [A] in boxes of side $L = 64$, the D1 plane in $L = 32$ boxes of the same patch, and the D1 and D2 intensities along the anti-diagonal $q = (t, 1-t, 0)$ at $L = 32$ and $L = 64$. Computed diffraction of a generated point set [C on patch]; not an observation, not a measured crystal, and not icosahedral. The fading D1 anti-diagonal is not a Bragg peak and not a diffuse sheet on this patch [X], and the figure makes no claim about the infinite hull; CH-V-020 stays [X]. Not a scored preregistered hypothesis. *Provenance:* quasicrystal-materials/diffraction/figures/chair44_L64_qz0_streak.png (sha256 `446f7757...`), copied unchanged to figs_v0.6/; data tables/chair44_L64_qz0_streak.json; preregistration sha256 `10253440...` (same patch).

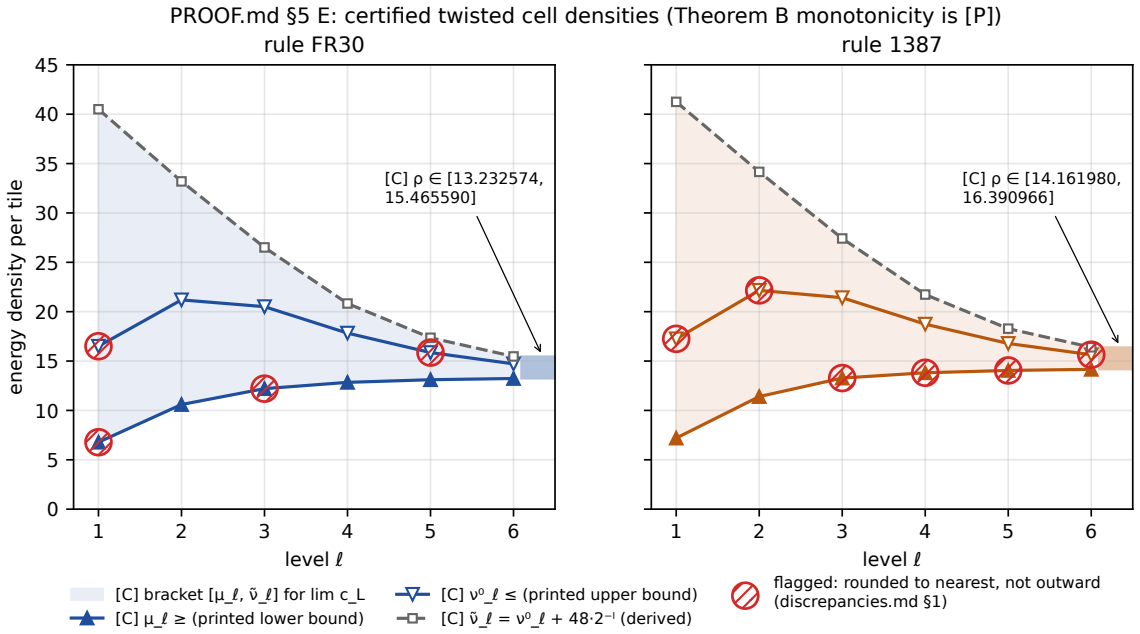


Figure 2: Certified twisted cell densities for FR30 (left) and 1387 (right), $\ell = 1$–6 (PROOF.md §5 E, l.173–182). Filled triangles: $\mu_\ell \geq$ [C]; open triangles: $\nu^0_\ell \leq$ [C]; grey squares: $\tilde{\nu}_\ell = \nu^0_\ell + 48 \cdot 2^{-\ell}$ [C]. The shaded band $[\mu_\ell, \tilde{\nu}_\ell]$ encloses $\lim c_L$ conditional on Theorems B, C, F [P, hand proof, unverified]. Right-edge bar: certified bracket for $\rho(d)$ [C], conditional in the same way. Red hatched rings: the 10 cells printed in PROOF.md rounded to nearest rather than outward; the outward values are in Table 1 (differences $\leq 5 \cdot 10^{-7}$, invisible at this scale). *Provenance:* figures/proof-brackets/figs_v0.2/fig1_mu_nu_vs_level.pdf (identical to frozen figs_v1.0); data/brackets_v1.0.json.

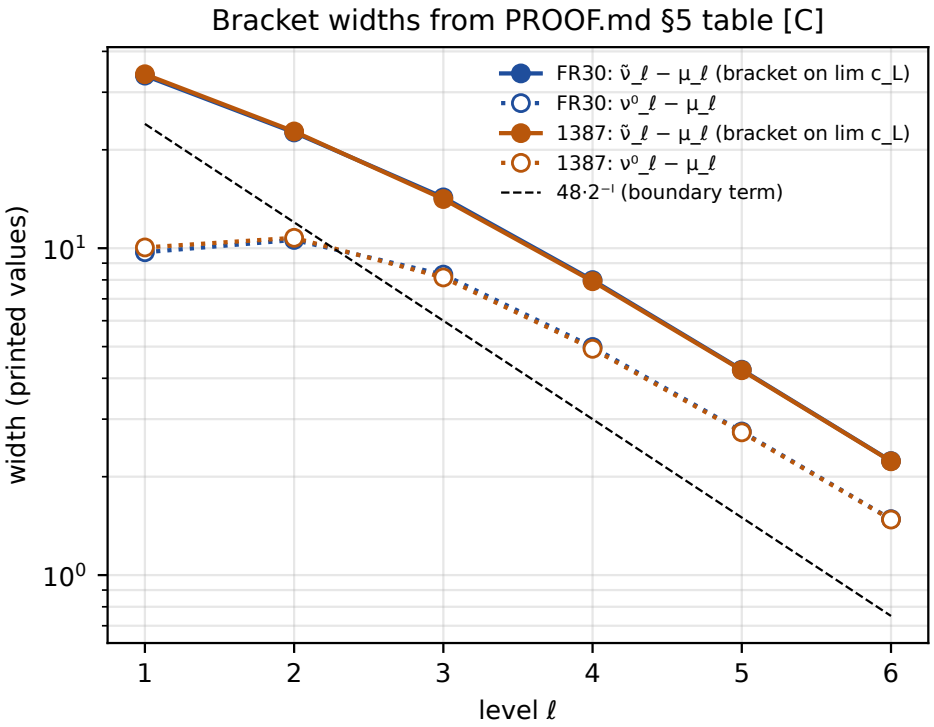


Figure 3: Bracket widths (log scale) from the printed PROOF.md §5 values [C]: $\tilde{\nu}_\ell - \mu_\ell$ (solid), $\nu^0_\ell - \mu_\ell$ (dotted), boundary term $48{\cdot}2^{-\ell}$ (dashed). At $\ell = 6$ the width is ≈ 2.23 for both rules. No trend beyond $\ell = 6$ is claimed. *Provenance:* figs_v0.2/fig2_bracket_widths.pdf (identical to frozen figs_v1.0).

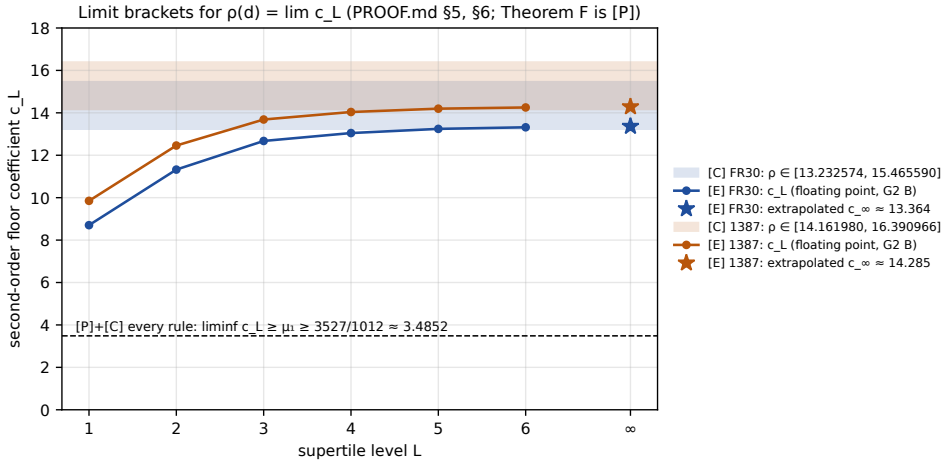


Figure 4: Limit brackets for $\rho(d) = \lim c_L$: $\rho(\mathrm{FR30}) \in [13.232574, 15.465590]$, $\rho(1387) \in [14.161980, 16.390966]$ [C], conditional on Theorems B, C, F [P, hand proof, unverified]. Dots: floating-point c_L, $L = 1$–6, from G2 Part B [E]; they have no error bars because they are deterministic solves (conjugate gradients, relative tolerance 10^{-12}, maximum relative residual about 10^{-12} or below). Stars: the preregistered B2 extrapolations $c_\infty \approx 13.364$, 14.285; an extrapolation beyond the data is printed here as [X] (the legend says [E]); changing the fitting range moves them by 0.10 and 0.04. Dashed: $\liminf c_L \geq 3527/1012 \approx 3.4852$ [C] (the legend says [P]+[C]; the paper prints the weaker component). The legend's "Theorem F is [P]" means a hand proof, unverified. Finite-L values are not bounded by the bands without the $\varepsilon_{\ell,L}$ factor of Theorem C. *Provenance:* figs_v0.2/fig3_limit_brackets.pdf (identical to frozen figs_v1.0); results/g2_partB.json.

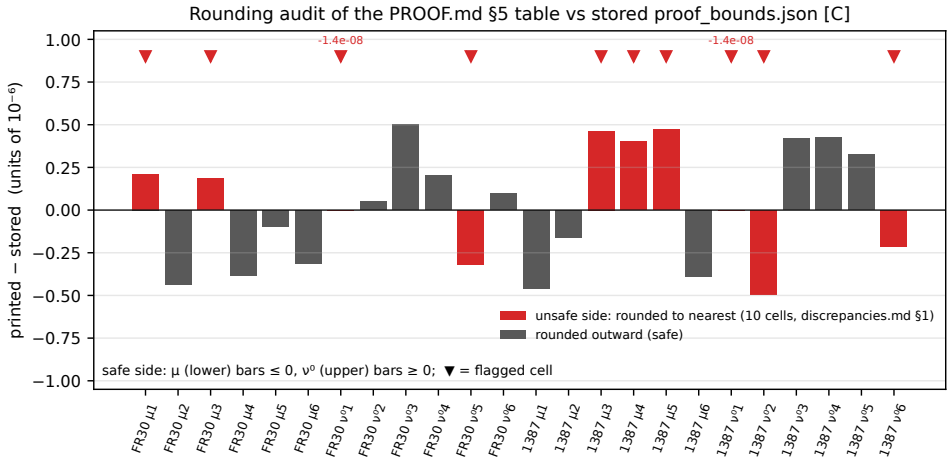


Figure 5: Rounding audit: printed PROOF.md value minus stored unrounded certified bound (results/proof_bounds.json), in units of 10^{-6}, all 24 cells [C]. Red bars (▼) are the 10 cells on the unsafe side; they match discrepancies.md §1. *Provenance:* figs_v0.2/fig4_rounding_audit.pdf (identical to frozen figs_v1.0).

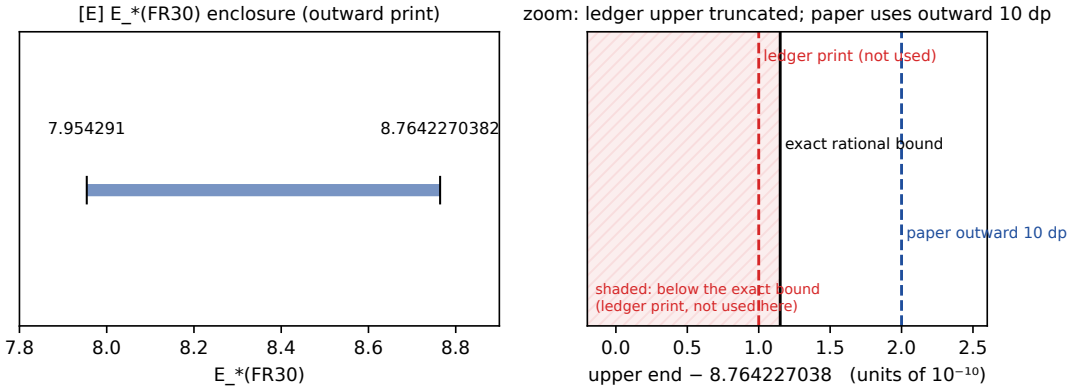


Figure 6: The FR30 spectral-floor enclosure as printed in this note, rounded outward: [7.954291, 8.7642270382] [E], until ratification of the Phase IX category-to-label mapping. The figure shows those outward values. The ledger prints [7.954292, 8.7642270381]; that upper end is truncated and lies below the exact rational bound, so it is not a safe upper print (right panel).[19] *Provenance:* figs_v0.2/fig5_Estar_FR30_enclosure.pdf; ledger.json l.474, l.624. Computed quantity, not observational.